



Smart Retail Inventory and Demand Prediction System

Group Name: INFO6007_Activity9_Ji_7

Course Name: INFO6007

AI Usage Declaration

Artificial intelligence tools, including ChatGPT, were used to support the preparation of this Project Management Plan. The use of AI was limited to assisting with brainstorming, structuring sections, improving academic writing, refining grammar, checking consistency across project management documents, and formatting references in Harvard style.

All project decisions, assumptions, project scope, schedule, governance structure, risk responses, quality controls, and final content were reviewed, evaluated, and edited by the group members. The group remained responsible for ensuring that the submitted work accurately reflected the selected project context, course requirements, lecture materials, and relevant academic and industry sources. AI-generated suggestions were not used without human review, and all cited sources were checked and incorporated by the group where appropriate.

Therefore, AI was used as a writing and editing support tool, rather than as a substitute for the group's own analysis, judgement, and project management decision-making.

Contribution Summary

Lihang Shen — 40%

- Completed the Risk Management Plan, Quality Management Plan, Communication Management Plan, Change Management Plan, and Project Monitoring and Control Plan.
- Took responsibility for final integration and consistency checking.
- Check the formatting for the whole paragraph.
- Revised inconsistent or low-quality sections from other members.
- Fixed the Stakeholder Management Plan and created the Power-Interest Grid.

Caresse Zhou — 40%

- Completed Project Scope Management Plan and Project Schedule Management Plan with good quality.
- Revise and improve sections written by other members.
- Check the formatting for the whole paragraph. Helped check consistency with the Project Charter.
- Actively supported the final editing, integration, and quality improvement.

Yiyang Li — 15%

- Provided first drafts of the Project Charter, Stakeholder Management Plan, Literature Review, and Project Governance and Roles. Later revisions were mainly improved by Zhou and Lihang Shen.
- Communication was limited.

Suhani Saha — 5%

- Provided initial drafts of the Project Cost Management Plan and Resource Management Plan with poor quality.
- Missed the internal deadline and rarely responded in WhatsApp.

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Project Charter

Topic	Smart Retail Inventory and Demand Prediction System		
Start Date	26 Feb 2026	End Date	23 June 2026
Brief Background			
The retail industry is currently under a fair amount of pressure from unstable supply chains out there, changing customer purchasing patterns, and also from competition coming from e-commerce platforms. For retailers, poor demand forecasting often leads to two major problems that cause trouble: one thing is stockouts of those popular products that customers want, and another thing is excessive inventory of slow-moving items that just sit on the shelves. Both of these issues tend to increase operating costs, reduce profit margins, and affect customer satisfaction in a negative way. As retail operations become more time-sensitive and more data-intensive over time, those traditional rule-based inventory management systems are starting to become less and less effective for handling these challenges. The target organization for this project is a mid-sized Australian retail chain that has 45 store locations in total. It currently manages its inventory through a reactive and rule-based kind of process, and this process is supported by separate ERP, WMS, and POS systems that don't talk to each other very well. Because these systems are not fully connected to each other, the organization ends up having limited visibility across its sales activities, its warehouse stock levels, and its procurement activities. This makes it pretty difficult for them to respond accurately to any changes in demand that happen. As a result of all this, the organization experiences what is known as a bullwhip effect, and what that means is small changes in customer demand can create much larger disruptions in ordering and stock control further up the chain. This situation has contributed to repeated stockouts of fast-moving consumer goods during those peak demand periods, as well as the build-up of dead stock in those lower-demand categories that don't move as quickly. To address these issues that have been identified, this project will develop and then deploy a Smart Retail Inventory and Demand Prediction System for the organization. The system will use machine learning models, and that includes LSTM and XGBoost, to generate SKU-level demand forecasts and to support automated procurement decisions where that makes sense. It will also connect with the organization's existing ERP, WMS, and POS systems so that sales information, inventory information, and purchasing information can all be used in a more coordinated way than they are right now. This will help the organization move away from relying mainly on past reports and move toward making more timely and forward-looking inventory decisions instead. The project is expected to improve inventory turnover rates, improve product availability, and also improve operational efficiency within about four months after commencement. It also supports the organization's broader digital transformation goals, and it does this by improving how inventory decisions are made across stores, warehouses, and procurement teams. By reducing stockouts and reducing excess inventory that is being held, the system will help the organization protect its profit margins and remain competitive in the Australian retail market going forward.			
Project Objectives			
• Design and deploy a cloud-native Artificial Intelligence (AI) forecasting engine utilising LSTM and XGBoost frameworks to predict SKU-level demand. • Establish secure, bi-directional API middleware to enable real-time inventory synchronisation across existing legacy POS, WMS, and ERP systems. • Develop a deterministic backend rules engine to automatically generate ERP-compliant Purchase Orders based on dynamic safety stock calculations.			

- Build a centralised, web-based operational dashboard featuring real-time inventory mapping and a Human-in-the-Loop (HITL) procurement console.
- Execute a comprehensive change management and training programme for all 45 Store Managers and central procurement staff prior to system deployment.

Project Success Criteria

- **Algorithm Performance:** Achieve a baseline demand forecast accuracy of $\geq 85\%$ (measured via MAPE) at the Store-SKU level within 90 days of Go-Live.
- **Operational Efficiency:** Attain a verifiable reduction in out-of-stock (OOS) incidents by $\geq 30\%$ compared to the pre-project baseline after 6 months.
- **Financial Impact:** Decrease excess inventory holding costs by $\geq 20\%$ within the 6-month post-deployment review period.
- **System Reliability:** Guarantee an API integration data loss rate of exactly zero, with an overall system uptime of $\geq 99.5\%$ post-deployment.
- **Organisational Adoption:** Secure a system usage and adoption rate of $\geq 90\%$ across the targeted operational staff within 60 days of launch.

Project Milestones

Milestone 1 — Project Initiation & Charter Approved: 04 March 2026
 Milestone 2 — Cloud & Data Infrastructure Established: 25 March 2026
 Milestone 3 — AI Model & API Middleware Complete: 29 April 2026
 Milestone 4 — System Integration Testing (SIT) Sign-off: 20 May 2026
 Milestone 5 — User Acceptance Testing (UAT) Sign-off: 03 June 2026
 Milestone 6 — System Go-Live & Staff Trained: 10 June 2026
 Milestone 7 — Project Closure (End of Hypercare): 23 June 2026

Budget Information

Total Approved Budget: AUD 500,000

- System Design & Architecture: AUD 60,000 (12%)
- AI/ML Model Development: AUD 120,000 (24%)
- System Integration & Development: AUD 100,000 (20%)
- Cloud Infrastructure & Hosting: AUD 80,000 (16%) (Note: Strict cap as per Scope baseline)
- Testing & Quality Assurance: AUD 50,000 (10%)
- Project Management, Training & Change Mgmt: AUD 40,000 (8%)
- Contingency Reserve: AUD 50,000 (10%)

Roles and Responsibilities

Name	Responsibilities	Role
Lihang Shen	Lead overall project planning, scheduling, budgeting, stakeholder communication, and risk management. Chair weekly team meetings and prepare progress reports.	Project Manager
Yiyang Li	Conduct requirements gathering, document business processes, coordinate UAT with end users, and manage stakeholder feedback.	Business Analyst
Qianyou Zhou	Design system architecture, develop AI/ML demand forecasting models, oversee technical integrations with ERP/POS/WMS systems.	Technical Lead / ML Engineer
Suhani Saha	Develop and execute test plans, perform QA, manage user training sessions, and support change management activities.	QA & Change Management Lead

Signature of Stakeholders

Lihang Shen
Yiyang Li
Qianyou Zhou
Suhani Saha
Finance Team
AI Team
Development Team
Retail Store Owner
Retail Store User representative

Stakeholder Management Plan

Introduction

Stakeholder management is an important part of delivering that Smart Retail Inventory and Demand Prediction System, and the reason for that is because the project involves quite a few different groups, and each of those groups comes with different responsibilities and different expectations. These groups include executive sponsors, the IT department, inventory and procurement staff, store managers, external vendors, and also system integration partners that are helping out. If these stakeholders are not clearly identified and properly managed from the start, the project could end up facing delays, unclear requirements, resistance from users, or disagreements over scope and priorities that are hard to resolve.

For this project, stakeholder management is being used to make sure the people and groups who will be affected by the system are involved at the right times and through the right communication channels that work for them. Early engagement is particularly important here, and that is because the new system is going to change how inventory decisions get made, how procurement activities get triggered, and how store-level stock information gets monitored day to day. Understanding what stakeholders expect at the very beginning of the project will help reduce resistance later on, improve the accuracy of the requirements we gather, and support a smoother adoption process during deployment. Previous research also highlights that active stakeholder engagement is closely linked to IT project success out there (Eskerod, Huemann and Savage, 2015).

This Stakeholder Management Plan goes on to explain how stakeholders will be identified, how they will be categorised, how they will be analysed, and how they will be engaged throughout the whole project lifecycle. The plan covers the period from project initiation and requirements gathering all the way through to deployment and that post-go-live review that happens at the end. The plan has been informed by stakeholder management research, including work by Jepsen and Eskerod (2009), and it will be reviewed at each major project milestone to make sure that stakeholder needs, stakeholder concerns, and their levels of influence all remain properly understood as the project moves forward.

Stakeholder Identification

Stakeholders for this project were identified through a structured kind of approach, and that approach combined a few different methods together. Those methods include project charter review, organizational chart analysis, expert interviews with department heads, and also some facilitated brainstorming sessions with the core project team that was assembled. This multi-method approach helps make sure we get comprehensive coverage of all parties who might affect the project or who might be affected by the project in some way.

Internal Stakeholders are those individuals and teams that sit inside the organization and are directly involved in the project, or directly impacted by the implementation of this new system. They include the Project Sponsor (who is the Head of Digital Transformation), the core project team (which includes the Project Manager, Business Analyst, Technical Lead, and QA Lead), the IT Department that handles infrastructure responsibilities, the Inventory and Procurement Team as the primary end-users, the Store Managers who will be interacting with the system on a daily basis, the Finance Department that oversees budget compliance for the project, and also the Data Analytics Team that will be supporting model validation work.

External Stakeholders on the other hand include parties that are outside the organization, but their involvement is still essential for project success. These external parties include retail suppliers and vendors who will need to interface with that automated replenishment engine, the cloud services provider (like AWS or Azure), the ML solution vendor, the ERP software vendor, and also some independent system integration consultants. Regulatory bodies that oversee data privacy compliance, and an example here would be compliance with the Australian Privacy Act, are also considered to be key external stakeholders that we have to keep in mind.

Internal Stakeholders	External Stakeholders
1. Project Sponsor (Head of Digital Transformation) 2. Project Manager 3. IT Department 4. Inventory & Procurement Team 5. Store Managers 6. Data Analytics Team 7. Finance Department	1. Retail Suppliers / Vendors 2. Cloud Services Provider (AWS/Azure) 3. ML Solution Vendor 4. ERP Software Vendor 5. System Integration Consultants 6. Regulatory Bodies (Australian Privacy Act) 7. End Customers (indirect beneficiaries)

Stakeholder Analysis

Each stakeholder has been assessed across three different dimensions that matter. The first dimension is power, and that refers to their ability to influence project decisions. The second dimension is interest, and that refers to their degree of concern in the project outcomes. The third dimension is impact, and that refers to the extent to which the project actually affects their work or their daily objectives.

The Project Sponsor holds high power and also high interest, and what this means is their continued support is really essential for securing those budget approvals and for resolving escalated issues when they come up. The IT Department and the Technical Lead possess high power as well, and that is because of their control over infrastructure and technical delivery, and they also have high interest since they are direct contributors to the work. Store Managers and the Inventory Team hold moderate power when you look at them, but they have very high interest because they are the primary end-users; any resistance coming from this group could seriously jeopardize adoption of the system.

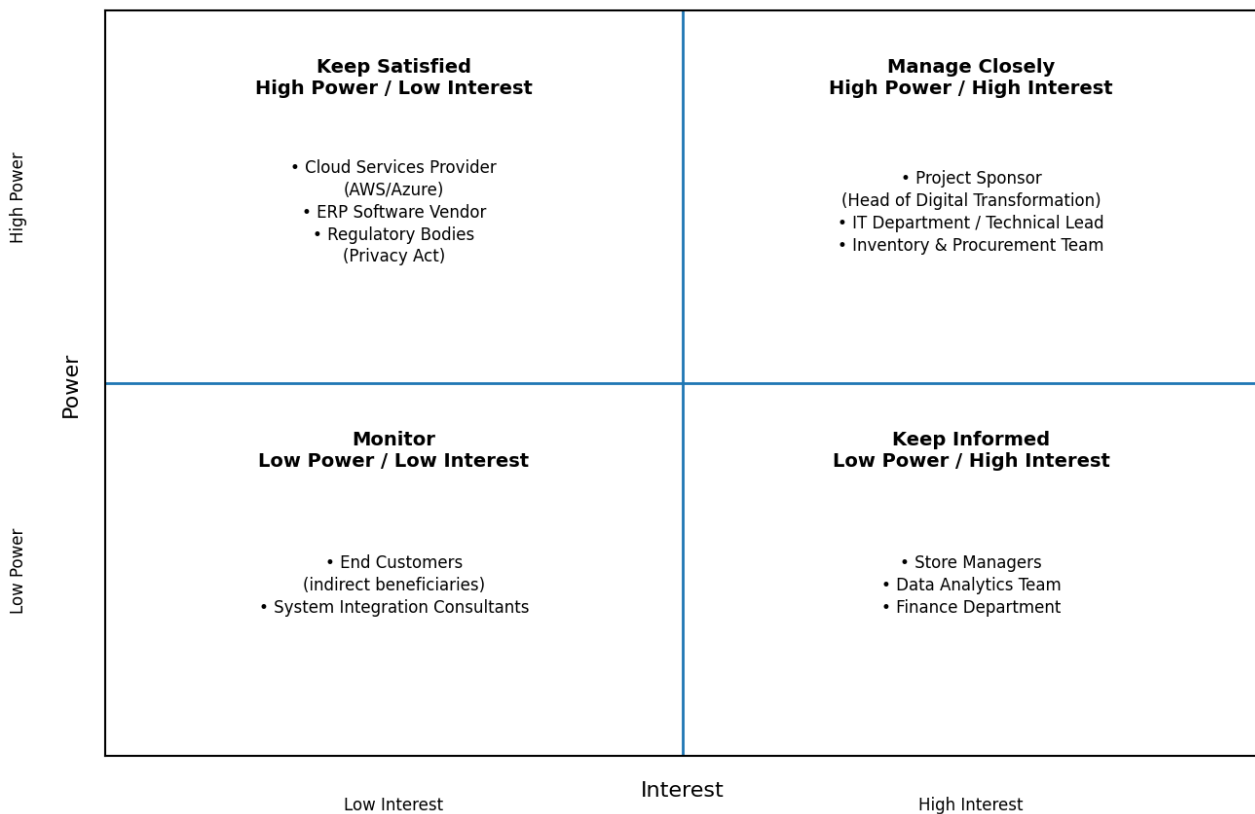
External stakeholders like the cloud services provider and the ERP vendor hold high power because they are technical dependencies for the project, but they have lower interest in the organizational outcomes that the project is trying to achieve. Retail suppliers have moderate interest and also moderate power, particularly in the context of that automated purchase order generation feature. Regulatory bodies have low day-to-day interest in the project, but they have high latent power, and the reason for that is non-compliance on our part could actually halt the project entirely.

Potential areas of resistance that we have identified include store managers who are concerned about operational disruption during the transition period, and also procurement staff who might be worried about role changes that could result from automation being introduced. These concerns have to be managed through targeted communication strategies and through those change management strategies that have been laid out in the plan.

Power Interest Grid

The Power-Interest Grid below categorises stakeholders into four quadrants to guide engagement strategies:

Power-Interest Grid for Stakeholder Engagement



HIGH POWER / HIGH INTEREST — Manage Closely:

- Project Sponsor (Head of Digital Transformation)
- IT Department / Technical Lead
- Inventory & Procurement Team

HIGH POWER / LOW INTEREST — Keep Satisfied:

- Cloud Services Provider (AWS/Azure)
- ERP Software Vendor
- Regulatory Bodies (Privacy Act)

LOW POWER / HIGH INTEREST — Keep Informed:

- Store Managers
- Data Analytics Team
- Finance Department

LOW POWER / LOW INTEREST — Monitor:

- End Customers (indirect beneficiaries)
- System Integration Consultants

Engagement Strategies

Engagement strategies have been tailored to each stakeholder group based on their position in the Power-Interest Grid:

- **Manage Closely (High Power, High Interest):** The Project Sponsor will receive weekly written status reports and participate in bi-weekly steering committee meetings. The IT Department and

core team will engage through daily stand-up meetings, shared project management tools (e.g., Jira/Confluence), and a dedicated project communication channel (Microsoft Teams). The Inventory and Procurement Team will be involved in requirements workshops and UAT sessions.

- **Keep Satisfied (High Power, Low Interest):** Vendors (cloud provider, ERP vendor) will be managed via formal contract reviews, SLA dashboards, and monthly performance meetings. Regulatory bodies will be engaged through compliance reports and legal counsel reviews at key project gates.
- **Keep Informed (Low Power, High Interest):** Store Managers will receive fortnightly project newsletters and participate in dedicated change management workshops before go-live. The Finance Department will receive monthly budget snapshots. The Data Analytics Team will be consulted during model validation phases.
- **Monitor (Low Power, Low Interest):** End customers and integration consultants will be monitored passively, with ad-hoc communication as needed. No formal engagement cadence is required for this group at this stage.

Communication Linkage

This Stakeholder Management Plan is directly linked to that Communication Management Plan we have, and the Communication Management Plan is what provides the detailed framework for handling all of the project communications that will happen. The communication preferences and the frequencies that have been defined above, and examples here include steering committee meetings, newsletters, dashboards, and workshops, are all formally documented and governed within that Communication Management Plan itself. All stakeholder communication that gets sent out will adhere to the information needs, the formats, and the escalation paths that have been defined in that plan, and doing this ensures consistency, traceability, and alignment across the whole project lifecycle. Any changes that come up to those stakeholder engagement approaches will be reviewed in accordance with the Change Management Plan, and then those changes will be reflected in both the Stakeholder Management Plan and the Communication Management Plan together.

Literature Review

Introduction

This literature review has been put together to support the development of a comprehensive Project Management Plan (PMP) for that Smart Retail Inventory and Demand Prediction System. The purpose of doing this review is to ground the plan in established theory and in contemporary practice, and it does this by synthesizing relevant academic literature and industry literature that are out there. The review examines three areas that are interconnected with each other: one area is the project management frameworks that are applicable to complex IT initiatives, another area is the specific management challenges that get posed by AI and machine learning (ML) technologies, and a third area is the identified gaps that exist in current practice, and these gaps directly inform a lot of the decisions that have been made throughout this PMP. The insights that have been drawn from the literature are then used to justify the methodological choices that appear across the governance section, the scope section, the schedule section, the risk management section, and the stakeholder management section of the document.

Literature and Industry Context

Project management of large-scale IT initiatives remains a persistent kind of challenge for organizations all around the world. Dwivedi et al. (2015) conducted a comprehensive review of IS project failure and success research, and what they confirmed was that the majority of IT initiatives continue to fall short of their original scope targets, time targets, or budget targets, and this is a pattern that has been documented consistently across different industries and different geographical regions. These findings really underscore just how important it is to apply rigorous, evidence-based project management frameworks when running these kinds of projects.

Gemino, Reich and Serrador (2021) examined 477 cross-industry projects and found that hybrid project management approaches, and these are approaches that combine the structured planning of traditional methodologies with the adaptive capacity of Agile, ended up achieving significantly higher stakeholder satisfaction when compared against purely traditional approaches, while still delivering comparable outcomes in terms of time, budget, and scope. Their findings directly inform the methodology that has been adopted in this PMP, and that methodology is a hybrid model that accommodates milestone-based governance to satisfy executive stakeholders on one hand, while still permitting iterative ML model development and testing cycles on the other hand.

Governance and stakeholder management emerge consistently throughout the literature as primary determinants of IT project success. Müller and Turner (2007) found that project governance structures, and that includes clearly defined roles, clearly defined decision rights, and clear escalation pathways, significantly reduce the likelihood of scope creep and budget overrun happening on a project. Flyvbjerg and Budzier (2011) further demonstrated in a study of 1,471 IT projects that one in six of those projects exceeded their budgets by an average of 200%, and this was often due to inadequate risk governance being in place. These findings directly motivate the robust governance structure, the RACI matrix, and the risk management processes that have been embedded into this PMP.

Furthermore, Fernandez and Fernandez (2008) argue that hybrid models are particularly well-suited to enterprise IT projects where the requirements are partially defined upfront but are still subject to change during development, and this is a characteristic that closely mirrors what we see with this retail AI initiative that we are working on.

Regarding technology-specific delivery outcomes, Feizabadi (2022) demonstrated that there are statistically significant supply chain performance improvements that come from machine learning-based demand forecasting when compared against traditional methods, while also noting that data quality and integration complexity are actually the primary barriers to operationalizing such systems in real-world practice. This finding directly informs the project's emphasis on ERP and POS integration planning and also on those structured change management activities that have been laid out.

Technology Considerations for Project Management

The Smart Retail Inventory and Demand Prediction System relies on several interdependent technologies out there, and those include machine learning forecasting models (like Long Short-Term Memory networks and gradient boosting algorithms such as XGBoost), real-time data pipelines, cloud-based infrastructure, and API-based integrations with the existing ERP and POS systems. While the technical design itself is addressed separately in other documents, these technology choices still carry some significant project management implications, and these implications show up across four key dimensions that matter.

The first dimension is Scope and Complexity. ML model development is inherently experimental in its nature. Unlike conventional software development projects, ML projects require iterative data collection, feature engineering, model training, and validation cycles before a production-ready model can actually be defined and locked down. Lwakatare et al. (2019) identify unique technical debt patterns and scope uncertainty in ML systems, and these things must be explicitly accounted for in the Work Breakdown Structure and in those scope change procedures. This really necessitates having a clearly defined scope boundary, one that separates model development iterations from the overall project timeline so they don't interfere with each other.

The second dimension is Risk and Uncertainty. AI projects tend to introduce non-deterministic risk into the planning, and what that means is model performance depends heavily on data quality, data volume, and data representativeness, and these things may not be fully known at the time of project initiation. Amershi et al. (2019) document that data-related challenges are actually the leading cause of AI project delays in enterprise settings. Because of this finding, a data readiness assessment has been incorporated as a formal early milestone right into the project schedule.

The third dimension is Resource and Skill Requirements. The project requires specialized ML engineering expertise, and this kind of expertise is distinct from the traditional IT project skills that most teams already have. Chuang et al. (2024) demonstrate that ML and AI technology creates significant skill gaps in organizations, with roles like ML engineers and AI data scientists requiring continuous development pathways that many organizations have not yet established for their staff. This creates some resourcing risks, and those risks have to be addressed properly in the Resource Management Plan that has been put together.

The fourth dimension is Schedule and Cost. Cloud infrastructure costs for those ML training workloads tend to be variable in nature, and they are also difficult to estimate accurately upfront before the work begins. Those GPU-accelerated compute instances on cloud platforms introduce some cost uncertainty into the budget, and that uncertainty requires having contingency budgeting in place and also requires close monitoring throughout the whole execution phase.

Knowledge Gaps and Project Implications

Despite the growing body of literature on AI project management, there are still several gaps that remain relevant to this particular initiative. One thing is most of the existing research on AI demand forecasting tends to focus on technical performance metrics, rather than looking at the project management processes that would be required to deliver such systems in complex organizational environments (Lwakatare et al., 2019). There is just limited guidance out there on how to structure governance for those hybrid AI and IT projects where the development cycles end up differing substantially from what you would see in traditional software delivery.

Another thing is while change management for ERP implementations has been well-studied over the years (Somers and Nelson, 2004), the literature on managing user adoption specifically for AI-driven retail tools remains pretty nascent at this stage. That behavioral resistance from operational staff, who may end up distrusting those algorithmic recommendations, represents a real challenge for which standardized mitigation frameworks are still underdeveloped and not widely available.

These gaps help inform two design decisions that have been made in this PMP. The first one is the adoption of a hybrid governance model, and this model is designed to accommodate both structured delivery milestones on one hand and iterative ML development sprints on the other hand. The second one is the inclusion of a dedicated change management stream within the project itself, and this stream comes with specific activities that target store manager adoption and procurement team adoption as their primary focus.

Summary

The literature that has been reviewed here establishes a pretty clear evidence base for all of those project management decisions that have been embedded into this PMP. Hybrid Agile-Waterfall delivery methods, and these are supported by Gemino et al. (2021), provide the methodological foundation for how the project will run. Lessons that were drawn from comparable ML-based supply chain deployments (Feizabadi, 2022) highlight just how critical data integration planning and change management really are, and both of those areas have been explicitly incorporated as separate workstreams in the plan. The governance structure and that RACI matrix both respond to the empirical evidence that links clear decision rights to IT project success, as shown by Müller and Turner (2007). Those technology-specific risks, particularly data readiness (Amershi et al., 2019), ML skill gaps (Chuang et al., 2024), and variable cloud costs, have been addressed in the Risk Management Plan and the Resource Management Plan respectively. When you put all of these insights together, they help make sure the project is grounded in both theoretical rigour on one hand and practical precedent on the other.

Project Governance and Roles

Introduction

Project governance in IT project management refers to that framework of policies, processes, roles, and decision-making structures that all work together to make sure a project gets delivered in an accountable, controlled, and transparent manner. Effective governance helps align project execution with the organizational strategy, makes sure resources are used efficiently across the board, and provides clear mechanisms for oversight, risk escalation, and issue resolution when problems come up.

For the Smart Retail Inventory and Demand Prediction System, having robust governance is particularly critical, and that is because of the project's technical complexity, its cross-functional dependencies, and the significant investment that has been made in it. Sirisomboonsuk et al. (2018) surveyed 533 IT professionals and found that both project governance and IT governance have a statistically significant positive impact on project performance, and this impact shows up specifically across those portfolio direction, project sponsorship, and disclosure and reporting dimensions. Without clearly defined accountability and decision rights in place, IT projects of this scale tend to be prone to scope creep, communication failures, and delayed delivery (Müller and Turner, 2007). This section therefore defines the project's governance structure, the roles and responsibilities of all the key participants, a RACI matrix for the major project activities, and the processes through which all decisions will be made, documented, and communicated throughout that whole project lifecycle.

Governance Structure

The project will operate under a three-tier governance structure that has been put in place. Tier 1 is Executive Oversight and this is handled by the Project Sponsor. The Project Sponsor, who is the Head of Digital Transformation, provides strategic direction for the project, gives final approval on any scope changes and budget changes that exceed AUD \$20,000, and handles escalation resolution when issues get passed up. The Sponsor chairs a monthly Steering Committee meeting, and that meeting is attended by the Project Manager and by key departmental heads from across the organization.

Tier 2 is Project Management and this is handled by the Project Manager. The Project Manager is responsible for day-to-day delivery, progress reporting, risk and issue management, and for coordinating across all of those functional workstreams that are running in parallel. The PM facilitates weekly team stand-up meetings and produces fortnightly status reports that go to the Sponsor for review.

Tier 3 is the Delivery Team, and this group includes the Technical Lead, the Business Analyst, and the QA Lead. The delivery team executes all of the technical work and analytical work, reports their status back to the Project Manager on a regular basis, and escalates any blockers they encounter through the defined channels that have been set up. Daily stand-ups are held within the technical team during those active development sprints when work is happening.

This hierarchy makes sure there are clear lines of accountability throughout the project, fast decision-making at the operational level where it matters most, and appropriate executive oversight for those high-impact decisions that need it. Decision escalation gets triggered automatically whenever issues cannot be resolved within 48 hours at that delivery team level.

Roles and Responsibilities

Project Sponsor (Head of Digital Transformation): This person provides executive sponsorship for the project, approves the project charter and the budget, champions the initiative at the board level, and makes the final decisions on any significant scope changes or budget changes that come up. They are accountable for making sure the project stays aligned with the organizational strategy.

Project Manager: This person is responsible for the overall planning work, schedule management, budget tracking, risk and issue management, stakeholder communication, and project reporting. They act as the single point of accountability for delivery against that Project Management Plan (PMP) we have in place.

Business Analyst: This person leads the requirements gathering activities, documents the user stories and functional specifications, facilitates workshops with those end-user groups, and coordinates the User Acceptance Testing (UAT) effort. They make sure the technical solution that gets built actually meets the business needs it was designed for.

Technical Lead / ML Engineer: This person is responsible for system architecture design, ML model development (which includes those demand forecasting algorithms), API integrations with the ERP, POS, and WMS systems, and also the cloud infrastructure setup. They act as the technical authority for all build decisions and integration decisions that get made.

QA & Change Management Lead: This person develops and executes the test strategy, and that includes unit testing, integration testing, and UAT. They manage defect tracking, coordinate the user training programmes, and implement the change management plan so that adoption can be supported properly. They also make sure the quality gates are met before each phase transition happens.

IT Department: This team provides infrastructure support, network access, security compliance reviews, and ongoing technical environment management for the project. They get consulted on any architectural decisions that have IT policy implications that need to be considered.

Inventory & Procurement Team: These people are the primary end-users of the system once it is live. They participate in requirements validation, UAT, and training sessions. They also provide subject matter expertise on the current inventory workflows and replenishment processes that are already in use.

RACI Matrix

The RACI matrix below defines accountability for all major project activities across the key project roles. R = Responsible (does the work), A = Accountable (owns the outcome), C = Consulted (provides input), I = Informed (receives updates). Each activity has exactly one Accountable owner.

Activity	Project Sponsor	Project Manager	Business Analyst	Technical Lead
Project Planning & Scheduling	A	R	C	C
Requirements Definition	I	A	R	C
ML Model Development	I	A	C	R
System Integration (ERP/POS/WMS)	I	A	C	R
Risk Management	I	R	C	C
Change Control Approval	A	R	C	C

Testing & UAT	I	A	C	C
User Training	I	A	C	I
Deployment & Go-Live	A	R	I	C
Budget Management	A	R	I	I
Stakeholder Reporting	I	R	I	I

Governance Process

All project decisions are classified into three different tiers, and this classification is based on how much impact they have on scope, cost, or schedule. Khalid et al. (2022) empirically demonstrated that inadequate project governance and IT governance are actually leading antecedents of schedule delay, and that finding reinforces the need for clearly defined decision thresholds and escalation pathways for the team to follow.

Tier 1 decisions are Operational Decisions and these fall under the Project Manager's authority. They are routine decisions that have no impact on the approved baseline. Examples of these would be task assignment changes, minor schedule adjustments that stay within a single sprint, and day-to-day resource allocation. These decisions get documented in the project log within 24 hours of being made.

Tier 2 decisions are Tactical Decisions and these require both the Project Manager and the Technical Lead. They are changes that affect technical design or schedule within a single workstream, and they cannot exceed a cost variation of AUD \$20,000. This type of decision requires written approval from the Project Manager, and it gets documented in the change log and then communicated to the Sponsor as well.

Tier 3 decisions are Strategic Decisions and these require Project Sponsor approval before anything can happen. This tier covers any change that affects the approved baseline by more than AUD \$20,000, any change that alters the project scope, or any change that shifts the go-live date by more than two weeks. A formal Change Request has to be submitted for these, then reviewed at the next Steering Committee meeting, and then approved by the Sponsor before any implementation work can start. All decisions across all tiers are recorded in that project decision register, and each entry includes the rationale behind the decision, the owner who made it, and the date it was made. Any issues that cannot be resolved within 48 hours get escalated up to the next governance tier for further handling.

Compliance

Project governance makes sure compliance is maintained across three different domains that matter. The first domain is Organizational Policies: the project will stick to the organization's IT governance policies, its procurement procedures, and its data management standards. All vendor contracts will be reviewed by the Legal team and the Procurement team before anyone signs them, and the technology choices that get made will be reviewed to check they align with that enterprise architecture roadmap. The governance framework adopted in this project aligns with the three-dimension model of project governance, and those dimensions are portfolio direction, project sponsorship, and disclosure and reporting. This model was validated by Sirisomboonsuk et al. (2018) as being positively associated with project performance outcomes.

The second domain is Data Privacy and Security: the system will handle sensitive retail transaction data, so compliance with the Australian Privacy Act 1988 (Cth) and the Australian Privacy Principles (APPs) is mandatory for the project. A Privacy Impact Assessment (PIA) will be conducted prior to the

system go-live date, and data encryption, access controls, and audit logging will all be implemented in accordance with the organization's information security policy. (Available at: <https://www.legislation.gov.au/Details/C2022C00235>, Accessed: 16 April 2026)

The third domain is Quality and Standards: the project will embed quality gate reviews at each major milestone, and the purpose of those reviews is to assess delivery against functional criteria, reliability criteria, usability criteria, and performance criteria. Governance checkpoints will be used to formally verify that each phase has met its defined acceptance criteria before the team is allowed to proceed to the next phase. Khalid et al. (2022) confirm that integrating governance controls at those phase transitions significantly reduces the likelihood of schedule deviations and quality deviations when you are dealing with complex IT projects.

Project Scope Management Plan

Project	Smart Retail Inventory and Demand Prediction System		
Start Date	26 Feb 2026	Prepared by	Caresse Zhou

Justification	The modern retail environment is characterized by macro-economic supply chain volatility, omnichannel complexity, and some pretty rapid shifts in consumer demand out there. The target organization, which is a mid-sized Australian retail chain that operates 45 locations, currently relies on a reactive and rule-based kind of inventory management, and it also relies on siloed legacy infrastructures like ERP, WMS, and POS systems. This fragmented approach creates a severe "bullwhip effect" for the company, and what that leads to is two critical financial pain points: one thing is chronic stockouts of fast-moving consumer goods (FMCG) during those peak demand periods, and another thing is the accumulation of dead stock that ends up locking up working capital and degrading the overall profit margins of the business. To secure its market position, the organization really needs a paradigm shift that moves from retrospective reporting over to predictive and more autonomous operations. The Smart Retail Inventory and Demand Prediction System justifies its capital investment by leveraging enterprise-grade Machine Learning, and examples of that would be gradient boosting frameworks like XGBoost and time-series neural networks like LSTM, so that it can process high-dimensional historical transactional data. By operationalizing these AI models through a scalable and secure cloud architecture (like AWS or Azure) and by establishing real-time API integrations with those existing core systems, the project will transform inventory procurement from a manual and heuristic-driven process into a highly precise and algorithmic capability. This initiative is really the cornerstone of the organization's digital transformation roadmap, and it has been designed to deliver a verifiable Return on Investment (ROI) by means of optimized inventory turnover, minimized holding costs, and maximized product availability across all locations.
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Objectives	The project objectives are defined across five fundamental pillars of delivery. These objectives adhere to SMART criteria and serve as the binding acceptance criteria for project closure and vendor sign-off. 1. AI/ML Modeling & MLOps Objectives 1.1 Forecast Accuracy Target: Design, train, and deploy a cloud-native Machine Learning forecasting engine capable of achieving a baseline forecast accuracy of $\geq 85\%$ at the Store-SKU level (measured via Mean Absolute Percentage Error - MAPE) within 90 days of system go-live. 1.2 Automated Retraining Pipeline: Implement an automated MLOps pipeline (e.g., AWS SageMaker pipelines) that triggers model retraining
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	bi-weekly using the latest POS data to prevent data drift and adapt to shifting seasonal trends. 1.3 Cold-Start & Edge-Case Handling: Develop explicit algorithmic heuristics to handle "cold-start" scenarios for newly introduced SKUs and highly sparse sales data items, ensuring 100% of the active catalogue receives a baseline forecast. 2. System Integration & Data Engineering Objectives 2.1 API Connectivity: Establish secure, bi-directional RESTful API endpoints between the new AI prediction engine and the legacy Enterprise Resource Planning (ERP), Point of Sale (POS), and Warehouse Management Systems (WMS). 2.2 Data Integrity Guarantee: Ensure zero critical data loss, duplication, or corruption during the historical data migration phase and ongoing real-time operational syncs. 2.3 Latency & Performance SLAs: Maintain an API response latency of <500ms for real-time inventory delta syncs during peak operational retail hours (9:00 AM – 6:00 PM). 2.4 Security & Compliance: Enforce end-to-end data encryption (AES-256 at rest, TLS 1.3 in transit) and implement role-based access control (RBAC) in strict compliance with the Australian Privacy Act 1988. 3. Application Frontend & Operational Dashboard Objectives 3.1 Enterprise UI Deployment: Develop and deploy a centralised, responsive Web User Interface (UI) accessible via standard enterprise web browsers for the 45 Store Managers and the central Procurement Team. 3.2 Real-Time Inventory Mapping: Deliver a dashboard module providing real-time visualisation of current stock levels across the central distribution centre and all physical retail nodes. 3.3 Predictive Alerting System: Implement a configurable push-notification system to flag projected stockouts or overstock risks with a minimum 7-day predictive lead time. 3.4 Human-in-the-Loop (HITL) Console: Build an override interface allowing senior procurement staff to review, manually modify, or approve ML-generated restock quantities prior to final execution. 4. Automated Replenishment & Execution Objectives 4.1 Dynamic Purchase Order Generation: Implement a deterministic rules-engine that consumes ML demand predictions to automatically
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	generate formatted Purchase Orders (POs) compliant with existing ERP standards. 4.2 Dynamic Safety Stock Calculation: Configure the engine to dynamically calculate and continuously adjust safety stock thresholds based on specific vendor lead times and predicted demand volatility. 4.3 Process Efficiency Gain: Reduce the manual processing time required for routine procurement and PO generation by a minimum of 80% compared to current operational baselines. 5. Business Value, Reliability & Adoption Objectives 5.1 Stockout Reduction: Attain a verifiable reduction in out-of-stock (OOS) incidents by $\geq 30\%$ against the FY2025 pre-project baseline, evaluated at the 6-month post-deployment milestone. 5.2 Holding Cost Reduction: Decrease excess inventory holding costs by $\geq 20\%$ within the same 6-month post-deployment review period. 5.3 High Availability SLA: Guarantee enterprise-grade system reliability with a measured Uptime of $\geq 99.5\%$, excluding explicitly approved and scheduled maintenance windows over the first three months. 5.4 Target User Adoption: Secure a system user adoption rate of $\geq 90\%$ across all target users within 60 days of launch, driven by a structured Change Management programme and mandatory training workshops.
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Summary of Project Deliverables	
PM Deliverables	Cloud Data Ingestion & ETL Pipeline - Functional Scope: Automated scripts responsible for extracting, transforming, and loading (ETL) historical transaction, promotion, and inventory data (from January 1, 2023, onwards) from legacy ERP and POS systems into a central cloud data lake. - Delivery Format & Platform: Python-based ETL scripts deployed via a cloud-native orchestration service (e.g., AWS Glue or Apache Airflow). Delivered as version-controlled source code (via enterprise GitHub/GitLab) along with executable deployment templates. - Application Scenario: Executes autonomously as a scheduled nightly batch job (e.g., at 2:00 AM) during non-peak retail hours to update the central repository without impacting daytime POS terminal performance. AI Demand Forecasting Module - Functional Scope: The core machine learning engine utilizing LSTM and XGBoost algorithms to predict SKU-level demand across all 45 store locations, achieving $\geq 85\%$ accuracy (MAPE).

	• Delivery Format & Platform: Serialised model artifacts (e.g., .pkl or ONNX formats) packaged within lightweight Docker containers. Deployed to a managed machine learning cloud environment (e.g., Amazon SageMaker) exposing secure inference endpoints. • Application Scenario: Polled dynamically by the central dashboard and replenishment engine every morning to generate a rolling 7-day forecast for the active catalogue of 15,000 SKUs. Automated MLOps Retraining Pipeline • Functional Scope: A continuous integration/continuous deployment (CI/CD) pipeline designed to automatically retrain the forecasting models using the latest transactional data to prevent data drift. • Delivery Format & Platform: Infrastructure-as-Code (IaC) configuration files (e.g., Terraform or AWS CloudFormation YAML files) and pipeline scripts. • Application Scenario: Runs as a background service on the 1st and 15th of every month. It trains a "challenger" model against the new data and automatically routes production traffic to the new model only if it mathematically outperforms the baseline "champion" model. Bi-Directional API Integration Middleware • Functional Scope: The critical communication layer that translates and routes data between the new cloud AI system and the legacy on-premise ERP (e.g., SAP/Microsoft Dynamics), WMS, and POS networks with zero critical data loss. • Delivery Format & Platform: A suite of RESTful APIs documented in OpenAPI 3.0 (Swagger) format, deployed as serverless functions (e.g., AWS Lambda) behind a secure API Gateway. • Application Scenario: Acts as a real-time bridge during business hours, ensuring that when a physical item is scanned at a POS terminal, the delta is securely transmitted (TLS 1.3 encrypted) to the cloud system within a latency threshold of <500ms. Automated Purchase Order (PO) Execution Engine • Functional Scope: A backend business rules engine that consumes the AI forecasts, applies vendor lead-time constraints and dynamic safety stock logic, and calculates replenishment quantities. • Delivery Format & Platform: Backend microservice application. The output delivery format of the POs will be strictly formatted data files (e.g., JSON, XML, or EDIFACT) that natively match the ingestion requirements of the existing legacy ERP system. • Application Scenario: Triggered daily at 4:00 PM, the engine automatically consolidates the day's predictions and pushes
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	formatted draft POs directly into the ERP's procurement module, effectively eliminating manual data entry. Web-Based Operational Dashboard & UI • Functional Scope: The centralised user interface featuring a real-time inventory map, predictive 7-day stockout alerts, and a Human-in-the-Loop (HITL) console for PO review and approval. • Delivery Format & Platform: A responsive Single Page Application (SPA) developed in React or Angular. Platform explicitly defined: This is a web-only application accessible via standard enterprise desktop browsers (Chrome 110+, Edge 110+) and responsive for enterprise-managed tablet browsers (iPadOS Safari). Note: Native mobile applications (iOS App Store / Android Google Play APKs) are strictly out of scope. • Application Scenario: Used daily by the 45 Store Managers via tablets while walking the retail floor to check stock statuses, and used concurrently by the central Procurement Team via desktop monitors for bulk PO approvals. Provisioned & Secured Cloud Infrastructure • Functional Scope: The foundational hosting environment required to run all aforementioned modules securely, guaranteeing $\geq 99.5\%$ uptime while adhering to the AUD \$80,000 budget cap. • Delivery Format & Platform: A fully operational enterprise cloud environment (e.g., AWS or Azure) encompassing Virtual Private Clouds (VPCs), Load Balancers, Identity and Access Management (IAM) roles, and GPU compute instances. Delivered with a comprehensive Architectural Topology Diagram (e.g., Visio/Lucidchart). • Application Scenario: Serves as the 24/7 production environment for the system, configured with AES-256 data-at-rest encryption to ensure strict compliance with the Australian Privacy Act 1988 for all processed retail data.
Product Deliverables	Comprehensive Project Management Plan (PMP) & Subsidiary Plans • Functional Scope: The master governance artifact that consolidates all planning dimensions of the project, including Scope, Schedule, Cost, Resource, Risk, Quality, Communication, and Change Management Plans. It establishes the formal baselines against which project performance will be measured. • Delivery Format & Platform: A formally formatted, consolidated PDF document conforming to academic and industry standards (e.g., PMBOK). Delivered via the organisation's enterprise document management system (e.g., SharePoint/Confluence). • Application Scenario: Serves as the binding execution contract. It requires a formal digital signature from the Project Sponsor (Head of Digital Transformation) prior to the commencement of the Execution phase.

Requirements Traceability Matrix (RTM)

- **Functional Scope:** A core quality assurance tool that bi-directionally maps every approved business requirement from the Inventory & Procurement Team to its corresponding technical architecture design, ML model feature, and final test script.
- **Delivery Format & Platform:** A dynamic, multi-tab enterprise spreadsheet (e.g., Microsoft Excel) or a configured requirements tracking board in Jira.
- **Application Scenario:** Utilised by the QA Lead during System Integration Testing (SIT) and User Acceptance Testing (UAT) to mathematically prove that 100% of the contracted features have been built and rigorously tested.

Live Governance Dashboards & Change Control Logs

- **Functional Scope:** The continuous tracking mechanisms for project health, encompassing the Risk Register, Issue Log, and the formal Change Control Log for recording all Tier 2 and Tier 3 strategic decisions.
- **Delivery Format & Platform:** Hosted live on a collaborative project management platform (e.g., Jira/Confluence). Change Control approvals will be documented as formal IT service tickets with timestamped audit trails.
- **Application Scenario:** Actively projected and reviewed by the Project Manager during weekly delivery team stand-ups and formally presented at the monthly Steering Committee meetings with the Project Sponsor.

Testing Strategy & UAT Sign-Off Certificates

- **Functional Scope:** Detailed documentation defining the testing methodology for the AI forecasting accuracy (targeting $\geq 85\%$), system integration (API latency $< 500\text{ms}$), and user workflows. It includes the final, legally binding sign-off sheets confirming the system is ready for production.
- **Delivery Format & Platform:** Formal testing procedure manuals (PDF) and digital sign-off certificates routed through an e-signature platform (e.g., DocuSign).
- **Application Scenario:** Executed at the conclusion of Milestone 5. The UAT certificates must be physically or digitally signed by the representative Subject Matter Experts from the Inventory Team and Store Managers to authorize the final Go-Live phase.

Change Management & User Training Asset Library

- **Functional Scope:** Educational and change-enablement materials specifically engineered to secure the target $\geq 90\%$ user adoption rate among operational staff, addressing potential behavioral resistance.
- **Delivery Format & Platform:** A comprehensive digital package including: two digital "Quick Reference Guides" (QRGs) in PDF

	format, interactive training modules (SCORM compliant for the corporate LMS), and presentation slide decks. - Application Scenario: Deployed directly to the 45 Store Managers and central Procurement Team during the scheduled half-day training workshops in mid-April 2026, acting as both teaching aids and long-term reference manuals. Go-Live Cutover Runbook & Post-Implementation Review (PIR) - Functional Scope: The Runbook details the minute-by-minute technical deployment steps, IT support rotas, and emergency rollback procedures for the system launch. The PIR is the final closure document evaluating overall project success, budget variance, and lessons learned. - Delivery Format & Platform: The Runbook is delivered as an actionable sequence spreadsheet. The PIR is delivered as a formal management report (Word/PDF) and an executive presentation deck. - Application Scenario: The Runbook is strictly followed by the Technical Lead and IT Department during the deployment weekend (Milestone 6). The PIR is presented by the Project Manager to the executive board at the formal Project Closure meeting.
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Project Scope	
In Scope	This section defines the strict, legally binding boundaries of the functional deliverables and operational services to be executed by the project team. Any activity, service, or feature not explicitly listed below is excluded from the project baseline. 1. Cloud Infrastructure & Security Provisioning Design, configuration, and provisioning of three distinct cloud environments (Development, UAT, and Production) on an enterprise cloud platform (e.g., AWS or Azure) using Infrastructure-as-Code (IaC), strictly capped at the allocated AUD \$80,000 budget. Configuration of network security, including Virtual Private Clouds (VPCs) and API firewalls, applying AES-256 (data-at-rest) and TLS 1.3 (data-in-transit) encryption to ensure explicit compliance with the Australian Privacy Act 1988. 2. Data Engineering & AI Model Development Development of automated ETL pipelines to extract, clean, and map historical transactional and inventory data sourced exclusively from January 1, 2023, to the present date.

	Engineering and training of machine learning algorithms (LSTM and XGBoost) to predict demand for the active catalogue of approximately 15,000 SKUs, targeting a baseline accuracy of $\geq 85\%$. Model Tuning Rework Limit: Algorithmic tuning is strictly limited to three (3) major hyperparameter optimization cycles during the development sprint to reach the 85% accuracy threshold. Development of an automated MLOps pipeline for bi-weekly model retraining in production. 3. Application & Middleware Engineering Development of a RESTful API gateway to serve as the bi-directional middleware enabling real-time data syncs (<500ms latency) between the cloud AI engine and the legacy on-premise ERP, WMS, and POS systems. Development of a web-based, responsive operational dashboard featuring a real-time inventory map, predictive stockout alerts, and a Human-in-the-Loop (HITL) procurement console. Development of an automated replenishment rules engine that computes safety stock and formats output data to match the native Purchase Order (PO) ingestion format of the existing ERP. UI/UX Rework Limit: The development of the web dashboard includes a maximum of two (2) rounds of client feedback and interface revision prior to the formal "Design Freeze" sign-off. 4. Testing, UAT Acceptance & Defect Rework Policy Execution of System Integration Testing (SIT) by the Technical Lead to verify API stability and data integrity with zero critical data loss. Facilitation of a strictly time-boxed fourteen (14) day User Acceptance Testing (UAT) window for the Inventory & Procurement Team and selected Store Managers. Acceptance & Rework Protocol: During UAT, the project team is responsible for remediating all identified Severity 1 (Critical System Failure) and Severity 2 (Core Functionality Impaired) defects. The scope includes a maximum of two (2) formal UAT bug-fix iteration deployment cycles. Scope Protection: Any user dissatisfaction arising from missing features that were not documented in the baseline Requirements Traceability Matrix (RTM) will not be treated as a "defect." Such requests will be rejected as Out of Scope and must be routed through the formal Tier 3 Change Request process for separate budget and schedule assessment.
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	5. Training, Go-Live, & Post-Deployment Support (Hypercare) Delivery of four (4) targeted, half-day functional training workshops for the 45 Store Managers and central procurement staff, alongside the provision of digital Quick Reference Guides (QRGs). Execution of the technical Go-Live cutover over a single weekend (Milestone 6). Hypercare Period (Post-Deployment Maintenance): Provision of a post-deployment Hypercare support period from 11 June 2026 to 23 June 2026 immediately following System Go-Live. During this window, the core project team will resolve any latent Severity 1 or 2 system bugs and monitor ML model stability at no additional project cost. System Handover: Delivery of technical architecture documentation, runbooks, and a formal knowledge transfer session to the internal IT Department to assume ongoing Tier 1 and Tier 2 operational support after the 30-day Hypercare period concludes.
Out of Scope	This section explicitly identifies the products, services, processes, and support mechanisms that are strictly excluded from the project baseline. Any request to incorporate the following items will be immediately rejected or must be processed as a formal Tier 3 Change Request requiring additional budget and timeline extensions. Hardware & Physical Infrastructure Procurement - The procurement, installation, maintenance, or configuration of any physical hardware. This explicitly excludes purchasing or upgrading Point-of-Sale (POS) terminals, warehouse barcode scanners, networking cables, routers, or on-premise physical servers for the 45 retail locations or central distribution centres. Legacy System Code Modification & Upgrades - Refactoring, patching, or modifying the internal core codebase, database schemas, or proprietary business logic of the legacy on-premise ERP, WMS, or POS platforms. The project team's responsibility is strictly limited to mapping to and calling the existing vendor-provided APIs. Any system upgrades required on the legacy side to support API connectivity are the sole responsibility of the organisation's IT Department or the respective software vendors. Historical Data Cleansing & Backfilling - Manual auditing, cleansing, data-entry correction, or back-filling of corrupted, missing, or unstructured historical data generated prior to January 1, 2023. The project assumes that the data extracted via existing systems from 2023 onwards is structurally valid. If historical data is missing or highly corrupted, the AI model will not mathematically interpolate it.

	Native Mobile Apps & Consumer-Facing Interfaces - The design or development of any native mobile applications (e.g., iOS App Store / Android Google Play APKs). The operational dashboard is explicitly defined as a web-based responsive application. Furthermore, the development of any e-commerce portals, customer loyalty apps, or direct-to-consumer tracking interfaces is strictly excluded. Extended System Support & Maintenance (Beyond Hypercare) - The provision of Tier 1, Tier 2, or Tier 3 IT helpdesk support, ongoing server maintenance, or manual machine learning model recalibration beyond the explicitly defined post-deployment Hypercare period ending on 23 June 2026. Following this Hypercare window, full operational ownership and maintenance accountability transfer completely to the organisation's internal IT Department. Cosmetic Rework & Out-of-SLA Defect Resolution - Remediation of Severity 3 (Minor) or Severity 4 (Cosmetic) defects during the User Acceptance Testing (UAT) phase if they do not actively impede core business functionalities or algorithm accuracy. Furthermore, any UI/UX modification requests submitted after the formal "Design Freeze" milestone, or exceeding the two (2) allotted revision cycles defined in the scope, will not be executed. Financial Processing & Logistics Execution - Automated financial clearing, vendor payment processing, or invoice settlement. While the automated engine will generate draft Purchase Orders (POs), the actual financial transaction and approval workflows remain entirely within the legacy ERP's finance modules. Additionally, physical supply chain optimisation, such as warehouse layout redesign or third-party freight routing, is excluded. Ongoing Cloud Consumption & Vendor Licensing Fees - The financial payment or ongoing assumption of third-party software licensing fees, ERP vendor API integration surcharges, or recurring cloud platform consumption costs (e.g., AWS monthly billing) occurring after the project's formal Go-Live and Closure dates. The AUD \$80,000 cloud budget applies exclusively to the development, testing, and deployment phases
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Project Assumptions
This section documents the foundational assumptions upon which the project's scope, schedule, and cost baselines have been constructed. Should any of these baseline assumptions be proven false during project execution, it will constitute a valid trigger for the Project Manager to initiate the formal Tier 3 Change Request process to reassess the schedule and budget.

1. Legacy System API Readiness & Vendor Cooperation

It is assumed that the organisation's existing legacy ERP (e.g., SAP/Microsoft Dynamics), WMS, and POS software possess standard, documented API connectivity (RESTful, GraphQL, or standard EDI/XML export capabilities) without requiring bespoke middleware to be built from scratch on the legacy side. Furthermore, it is assumed that the legacy software vendors will provide access to accurate API documentation and testing sandbox environments within fourteen (14) days of project kick-off.

2. Historical Data Quality & Structural Integrity

It is assumed that the historical transactional, promotional, and inventory data generated from January 1, 2023, onwards is structurally sound, continuous, and representative of actual retail operations. The project schedule and budget assume that the ML models can achieve the targeted $\geq 85\%$ accuracy without necessitating a massive, unbudgeted manual data-cleansing or back-filling phase to correct corrupt legacy records.

3. Algorithmic Feasibility & Demand Predictability

It is assumed that the consumer purchasing behaviour across the 15,000 SKUs exhibits mathematically identifiable patterns (seasonality, trend, or cyclicity). The targeted $\geq 85\%$ SKU-level forecast accuracy is assumed to be inherently achievable via industry-standard ML algorithms (LSTM, XGBoost) based on the quality of the provided POS data.

4. Cloud Cost Stability & Resource Provisioning

It is assumed that the pricing models for the selected enterprise cloud provider (e.g., AWS or Azure) will remain stable throughout the project lifecycle. Specifically, the variable costs associated with on-demand GPU-accelerated compute instances required for iterative Machine Learning model training will not experience unpredictable price surges and will be fully accommodated within the AUD \$80,000 cloud infrastructure budget.

5. Store Network Infrastructure & Bandwidth

It is assumed that all 45 physical retail store locations and the central distribution centres currently possess stable, commercial-grade internet infrastructure. The existing network bandwidth is assumed to be sufficient to support real-time POS data synchronisation to the central cloud platform without exceeding the $< 500\text{ms}$ latency threshold during peak retail operating hours.

6. Stakeholder Engagement & SME Availability

It is assumed that all identified Key Stakeholders, specifically the primary Subject Matter Experts (SMEs) from the Inventory & Procurement Team and selected Store Managers, will be made available to participate in scheduled requirements gathering workshops, User Acceptance Testing (UAT), and training sessions. It is assumed they will allocate a minimum of 6 hours per week during critical phase-gate milestones without reprioritising their operational duties over project commitments.

7. Executive Enforcement & Adoption Mandate

It is assumed that the Project Sponsor (Head of Digital Transformation) possesses the organisational authority and executive willingness to mandate the use of the new system. It is assumed that executive management will actively enforce the system's adoption to mitigate potential behavioural resistance from operational staff, ensuring the $\geq 90\%$ adoption target can be met within 60 days.

8. Regulatory & Compliance Stability

It is assumed that there will be no immediate, retroactive legislative amendments to the Australian Privacy Act 1988 (Cth) or Australian Data Sovereignty regulations during the 4-month project execution window (Feb - June 2026) that would necessitate a mid-development architectural redesign of the data encryption or cloud storage protocols.

9. Core Team Retention & Skill Adequacy

It is assumed that the designated core project delivery team (Project Manager, Business Analyst, Technical Lead/ML Engineer, and QA/Change Management Lead) will remain intact for the entirety of the project lifecycle. It is also assumed that the existing enterprise IT Department possesses the baseline technical competency required to assume Tier 1 and Tier 2 support handover upon the conclusion of the 30-day Post-Deployment Hypercare period.

Project Schedule Management Plan

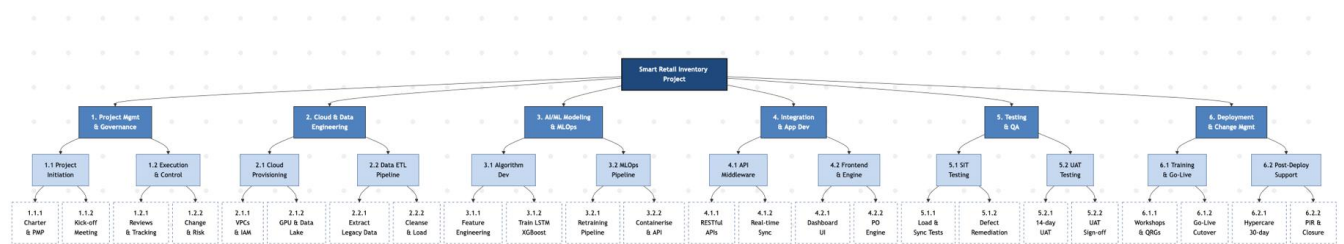
Introduction

Schedule management in complex enterprise IT projects, and particularly in those projects that involve predictive Artificial Intelligence (AI) and legacy system integrations, is really the definitive process of planning, developing, managing, and controlling the project timeline so that punctual delivery can be ensured. In the context of that Smart Retail Inventory and Demand Prediction System, a rigorous Schedule Management Plan is critical for addressing the inherent uncertainties that come with machine learning model training and also for mitigating those schedule risks that are associated with cross-functional dependencies, like API readiness from those legacy ERP vendors that the organization works with. This plan establishes a definitive temporal baseline, and the purpose of that baseline is to coordinate the efforts of the technical delivery team, align the expectations of executive stakeholders, and guarantee that the system can achieve its operational go-live target without compromising testing rigor or data integrity along the way.

By adopting a hybrid scheduling methodology that uses rigid Waterfall milestones for executive governance on one hand and iterative Agile sprints for ML algorithm development on the other hand, this document outlines the explicit processes for developing and controlling the project schedule. It details the Work Breakdown Structure (WBS), maps out the sequential task dependencies by means of Network Diagramming, applies the Program Evaluation and Review Technique (PERT) for probabilistic duration estimation, and also identifies the core Critical Path for the project. Ultimately, this framework makes sure that all of the project activities stay synchronized, that the technical resource allocations get optimized where possible, and that any potential schedule variances are proactively managed before they become bigger problems.

Schedule Development Process

The Work Breakdown Structure

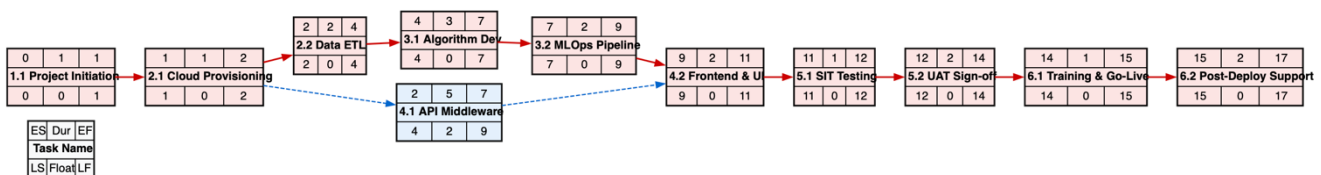


WBS Item No	Description	Dependencies (if any)
1	Project Management & Governance	
1.1	Project Initiation: Establish project baseline, team, and initial documentation.	
1.1.1	Draft and approve Project Charter and Project Management Plan (PMP).	None
1.1.2	Conduct formal Project Kick-off meeting with the Steering Committee.	1.1.1

1.2	Project Execution & Control: Ongoing tracking of schedule, budget, and risks.	
1.2.1	Facilitate weekly stand-ups and monthly Steering Committee reviews.	1.1.2
1.2.2	Manage Tier 2/Tier 3 Change Requests and update the Risk Register.	1.1.2
2	Cloud Infrastructure & Data Engineering	
2.1	Cloud Environment Provisioning: Set up AWS/Azure resources.	
2.1.1	Configure Virtual Private Clouds (VPCs), IAM roles, and firewalls.	1.1.2
2.1.2	Provision GPU-accelerated compute instances and data lake storage.	2.1.1
2.2	Data ETL Pipeline Development: Build historical data ingestion scripts.	
2.2.1	Extract historical POS, WMS, and ERP data (Jan 2023 onwards).	2.1.2
2.2.2	Cleanse, transform, and load data into the secure cloud data lake.	2.2.1
3	AI/ML Modeling & MLOps	
3.1	Algorithm Development: Train the core demand forecasting models.	
3.1.1	Perform feature engineering on ingested historical retail data.	2.2.2
3.1.2	Train and tune LSTM and XGBoost models to achieve $\geq 85\%$ accuracy.	3.1.1
3.2	MLOps Pipeline Integration: Automate the model lifecycle.	
3.2.1	Develop and deploy the bi-weekly automated model retraining pipeline.	3.1.2
3.2.2	Containerise approved models (e.g., Docker) and expose inference endpoints.	3.1.2
4	System Integration & Application Development	
4.1	API Middleware Development: Build the bi-directional communication bridge.	
4.1.1	Develop RESTful APIs connecting the AI engine to legacy ERP/WMS/POS.	2.1.1
4.1.2	Configure real-time data syncs ensuring <500ms latency and TLS 1.3 encryption.	4.1.1
4.2	Frontend & Replenishment Engine: Build the UI and business logic.	
4.2.1	Develop the Web-Based Operational Dashboard (inventory map, alerts, HITL).	4.1.1
4.2.2	Code the automated rules engine to generate formatted Purchase Orders.	3.2.2, 4.1.1
5	Testing & Quality Assurance	
5.1	System Integration Testing (SIT): Validate technical architecture.	
5.1.1	Execute API load testing and verify zero critical data loss during syncs.	4.1.2, 4.2.2
5.1.2	Remediate any Severity 1 or 2 defects identified during SIT.	5.1.1
5.2	User Acceptance Testing (UAT): Validate business requirements.	

5.2.1	Facilitate 14-day UAT with the Procurement Team and Store Managers.	5.1.2
5.2.2	Secure formal UAT sign-off certificates from key stakeholders.	5.2.1
6	Deployment & Change Management	
6.1	Training & Go-Live: Execute user enablement and system launch.	
6.1.1	Deliver half-day training workshops and distribute Quick Reference Guides.	5.2.2
6.1.2	Execute Go-Live cutover runbook and deploy system to production.	6.1.1
6.2	Post-Deployment Support: Project closure and handover.	
6.2.1	Provide 15-days Hypercare support to resolve latent production bugs.	6.1.2
6.2.2	Deliver Post-Implementation Review (PIR) and formally close project.	6.2.1

Network Diagram



The Network Diagram illustrates the logical sequencing and the interdependencies of all those Level 2 project activities, and it does this by using the Precedence Diagramming Method, or PDM. The critical path, and that critical path represents the longest sequence of dependent tasks that have zero float attached to them, ends up dictating the shortest possible project duration, which comes out to 17 weeks in total. A strategic architectural decision was made to execute Track A (which is the Data and AI/ML Modeling track) and Track B (which is the API Middleware Development track) at the same time, concurrently with each other. Because the API middleware development work requires less time when compared against that highly iterative ML modeling phase, the middleware track inherently possesses about two weeks of schedule float built into it. This deliberate parallel routing helps optimize resource allocation, and what that means is it allows the Technical Lead to shift focus between the two tracks whenever needed without jeopardizing that final integration point at the Application Development phase.

Estimating resources for Activities

WBS Item No	Resources
1.1	Human: Project Sponsor, Project Manager. Software: Confluence, MS Word, DocuSign (for charter approval).
1.2	Human: Project Manager. Software: Jira (for tracking), MS Teams (communications), MS Project (scheduling).
2.1	Human: Technical Lead, IT Department.

	Technical/Hardware: AWS/Azure Cloud Subscriptions, VPC, IAM, GPU Compute Instances. Software: Terraform (Infrastructure-as-Code).
2.2	Human: Technical Lead/ML Engineer, Data Analytics Team. Technical: Cloud Data Lake Storage (e.g., AWS S3). Software: Python, Apache Airflow / AWS Glue.
3.1	Human: Technical Lead/ML Engineer, Data Analytics Team. Technical: GPU Compute Instances (for heavy model training). Software: Jupyter Notebooks, XGBoost/LSTM libraries, Python.
3.2	Human: Technical Lead/ML Engineer. Technical: Cloud ML Environment (e.g., AWS SageMaker). Software: Docker, GitHub/GitLab (for CI/CD version control).
4.1	Human: Technical Lead/ML Engineer, IT Department. Technical: API Gateway Serverless compute. Software: Swagger (OpenAPI 3.0), Postman (for API dev).
4.2	Human: Business Analyst, Technical Lead/ML Engineer. Software: React/Angular (Frontend UI), Node.js/Python (Backend Engine), IDEs (VS Code).
5.1	Human: QA Lead, Technical Lead/ML Engineer. Software: Selenium (UI testing), JMeter (API load testing), Jira (Defect tracking).
5.2	Human: QA Lead, Business Analyst, Inventory Team SMEs, Store Managers. Technical: UAT Cloud Environment (Staging). Software: Jira (Defect logging), DocuSign (UAT Sign-off).
6.1	Human: QA & Change Management Lead, Project Manager, Store Managers. Software: Corporate LMS (Learning Management System), PDF Readers (for QRGs), Production Cloud Environment.

Estimating durations

To establish a realistic and mathematically sound project timeline, the duration for each Level 2 activity was calculated using the Program Evaluation and Review Technique (PERT). Because machine learning development and legacy system integration carry inherent uncertainties, PERT provides a weighted average duration rather than a single-point estimate.

The duration in weeks (E) was calculated using the standard PERT formula:

$$E = \frac{O + 4M + P}{6}$$

(Where O = Optimistic estimate, M = Most likely estimate, and P = Pessimistic estimate).

For example, for WBS 3.1 (Algorithm Development), the estimates were: Optimistic (2 weeks), Most Likely (3 weeks), Pessimistic (4 weeks). Using the formula, the expected duration resolves to exactly 3 weeks.

Note: WBS 1.2 (Project Execution & Control) is a Level of Effort (LOE) activity that spans the entire lifecycle of the project, hence its duration encompasses the full 17 weeks.

WBS Item No.	Start Date	End Date	Duration (weeks)
1.1	26 Feb 2026	04 Mar 2026	1
1.2	26 Feb 2026	23 Jun 2026	17
2.1	05 Mar 2026	11 Mar 2026	1
2.2	12 Mar 2026	25 Mar 2026	2
3.1	26 Mar 2026	15 Apr 2026	3
3.2	16 Apr 2026	29 Apr 2026	2
4.1	12 Mar 2026	15 Apr 2026	5
4.2	30 Apr 2026	13 May 2026	2
5.1	14 May 2026	20 May 2026	1
5.2	21 May 2026	03 Jun 2026	2
6.1	04 Jun 2026	10 Jun 2026	1
6.2	11 Jun 2026	23 Jun 2026	2

Critical Path Method

Based on the precedence logic defined in the Network Diagram and the PERT duration estimates, the Critical Path Method (CPM) was applied to identify the longest continuous sequence of activities from project initiation to project closure. This path determines the shortest possible time in which the project can be completed (17 weeks). Any delay in a critical activity will directly delay the final Go-Live and project closure dates.

1. The Critical Path The mathematical Critical Path for this project is formally identified as: **Start → 1.1 → 2.1 → 2.2 → 3.1 → 3.2 → 4.2 → 5.1 → 5.2 → 6.1 → 6.2 → End**

2. Critical Activities (Zero Slack / Zero Float) The activities listed below lie directly on the critical path. They have zero weeks of total float/slack, meaning their early start/finish times equal their late start/finish times. Close monitoring of these tasks is paramount:

- 1.1 Project Initiation
- 2.1 Cloud Environment Provisioning
- 2.2 Data ETL Pipeline Development
- 3.1 Algorithm Development

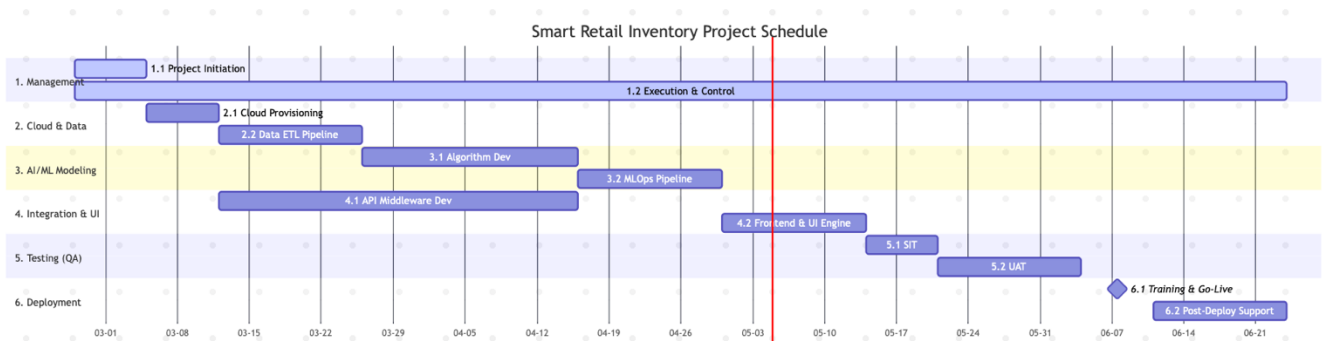
- 3.2 MLOps Pipeline Integration
- 4.2 Frontend & Replenishment Engine
- 5.1 System Integration Testing (SIT)
- 5.2 User Acceptance Testing (UAT)
- 6.1 Training & Go-Live
- 6.2 Post-Deployment Support

3. Non-Critical Activities and Slack/Float Analysis To optimise resource utilisation, **WBS 4.1 (API Middleware Development)** is deliberately scheduled to run in parallel with the data and machine learning tracks (WBS 2.2, 3.1, and 3.2).

- **Path A (Data/ML Track - Critical):** WBS 2.2 (2w) + 3.1 (3w) + 3.2 (2w) = **7 weeks total duration**.
- **Path B (API Track - Non-Critical):** WBS 4.1 = **5 weeks total duration**.
- **Float/Slack Identification:** Since Path A and Path B converge before WBS 4.2 can begin, **WBS 4.1 possesses a Total Slack/Float of exactly two (2) weeks**. This strategic buffer means that if unexpected complexities arise while integrating with the legacy ERP/POS APIs, the middleware development team has a 2-week safety margin before their delays begin to impact the overall project go-live date.

(Note: WBS 1.2 Project Execution & Control is an overarching support activity with a fixed duration bound to the project timeline, and thus carries zero float.)

Gantt Chart



The Gantt Chart translates that network logic into a chronological project timeline, and this timeline spans from project initiation on 26 February 2026 all the way to formal closure on 23 June 2026. This schedule arrangement has been deliberately structured using a Hybrid Agile-Waterfall methodology, and the purpose of that is to mitigate some specific technology delivery risks that we have identified. Sequential and fixed Waterfall milestones are enforced for the initial Cloud Provisioning phase and also for the final UAT and Go-Live phases, and the reason for that is to ensure strict executive governance, security compliance, and operational stability throughout those parts.

Conversely, the AI/ML Modeling phase, the API Integration phase, and the Frontend UI phase have been purposefully arranged so that they overlap with each other. This schedule compression technique is known as fast-tracking, and what it does is it allows the engineering team to conduct iterative Agile development sprints. By testing API endpoints and tuning the ML models at the same time, the team can

avoid waiting for one distinct phase to finish entirely before they are allowed to start the next phase of work.

Furthermore, the schedule deliberately sets that System Go-Live date to occur by early June. This strategy intentionally reserves the final segment of the project timeline, and that segment is set aside exclusively for a dedicated Post-Deployment Support window, which is also called the Hypercare period. Doing this ensures that any latent production defects that show up can be resolved prior to that absolute project closure deadline at the end of June.

Review of the Schedule

To ensure the viability, accuracy, and operational feasibility of the project timeline, a comprehensive, multi-step schedule review was conducted prior to baselining the schedule. The review process actively involved the Project Manager, Technical Lead, and key stakeholders to validate the logical flow of activities against the constrained 17-week timeframe (26 Feb 2026 – 23 Jun 2026). The specific actions taken to review and validate the schedule were as follows:

- 1. WBS Completeness & Scope Alignment Check:** The schedule was cross-referenced against the Scope Management Plan and the WBS Dictionary. We verified the 100% rule to ensure that all Level 2 and Level 3 deliverables were fully represented as distinct schedule activities with no missing work packages. This verification specifically included the critical machine learning model training, API integrations, and UAT phases.
- 2. Dependency & Network Logic Verification:** A forward and backward pass analysis was conducted on the Precedence Diagramming Method (PDM) network diagram. This confirmed that all Finish-to-Start (FS) dependencies were logical. Specifically, we verified that the convergence point (WBS 4.2 Frontend & Replenishment Engine) correctly waits for both Track A (MLOps) and Track B (API Middleware) to finish before UI integration can commence.
- 3. Resource Feasibility & Leveling Analysis:** We cross-checked the *Estimating resources for Activities* matrix against the parallel timeline. Because the Technical Lead is required on both the Data/ML track and the API Middleware track simultaneously, we verified that the calculated 2-week float/slack on the API track allows for adequate "Resource Leveling." This ensures the Technical Lead is not mathematically overallocated (>100% capacity) during weeks 3 to 7.
- 4. PERT Estimate Validation:** The Optimistic, Pessimistic, and Most Likely duration estimates used in the PERT calculations were reviewed with the Technical Lead. We specifically ensured that sufficient buffer was built into WBS 3.1 (Algorithm Development) to account for the iterative nature of ML hyperparameter tuning.
- 5. Milestone & Objective Alignment Check:** Finally, the calculated Critical Path dates were mapped against the strategic business objectives. We confirmed that the scheduled Go-Live (10 June 2026) safely precedes the final project closure deadline (23 June 2026), leaving a robust 2-week window exclusively for the required Post-Deployment Hypercare support

Milestones

The following table details the major project milestones, marking the completion of critical phases and the delivery of key phase-gate artifacts. These dates are derived directly from the Critical Path Method calculation and serve as the primary progress tracking points for the Steering Committee.

Sr No.	Milestone Name	Description	Date of achievement
1	Project Initiation & Charter Approved	Formal approval of the Project Management Plan (PMP) and Project Charter by the Project Sponsor. Marks the end of WBS 1.1.	04 Mar 2026
2	Cloud & Data Infrastructure Established	Successful provisioning of the AWS/Azure environment and completion of the automated ETL data ingestion pipeline (WBS 2.2).	25 Mar 2026
3	AI Model & API Middleware Complete	Convergence of parallel development tracks. ML models are trained ($\geq 85\%$ accuracy) and legacy API middleware is fully developed (WBS 3.2 & 4.1).	29 Apr 2026
4	System Integration Testing (SIT) Sign-off	Frontend dashboard, automated PO engine, and backend models are integrated. Technical Lead signs off on zero critical data loss and <500ms API latency (WBS 5.1).	20 May 2026
5	User Acceptance Testing (UAT) Sign-off	Completion of the 14-day testing window. The Inventory & Procurement Team formally signs UAT certificates, authorizing production deployment (WBS 5.2).	03 Jun 2026
6	System Go-Live & Staff Trained	Successful execution of the Go-Live cutover runbook over the weekend. Target user base has completed all necessary change management training (WBS 6.1).	10 Jun 2026
7	Project Closure & Handover	Conclusion of the post-deployment Hypercare window. Delivery of the Post-Implementation Review (PIR) and formal handover to IT support. (WBS 6.2).	23 Jun 2026

Cost Management Plan

Introduction

Project cost management is the process of planning, estimating, budgeting, financing, monitoring, and controlling all of the project costs, and the purpose of this is to make sure the project can be completed within that approved budget that has been set. For IT projects, effective cost management is particularly important because the project expenditure often ends up being affected by specialized human resources, cloud infrastructure costs, software tools, vendor dependencies, system integration complexity, cybersecurity requirements, and also all of the testing activities that need to happen.

For the Smart Retail Inventory and Demand Prediction System, cost management is essential because the project involves multiple high-cost workstreams, and those include AI/ML model development, cloud infrastructure provisioning, API middleware development, ERP and POS and WMS integration work, system testing, user training, and change management activities. Without a structured cost management process in place, the project could experience budget overruns, uncontrolled cloud consumption, unexpected integration costs, or even insufficient funding for testing and deployment when those phases arrive.

This Cost Management Plan defines how the project costs will be estimated, approved, allocated, monitored, and controlled throughout that whole project lifecycle. The total approved budget, just like what was stated in the Project Charter, is AUD 500,000. This budget includes all of the planned project expenditure and also the contingency reserve that has been set aside. The plan therefore establishes the cost baseline, and that baseline is what the actual spending will be measured against. It also defines how the funding will be released over time, how any cost variance will be reviewed when it appears, and how budget changes will be managed through that project governance and change control process we have in place.

Cost estimation approach

The project will use a bottom-up cost estimation approach for its planning. What this means is the cost of each major workstream will be estimated separately, and then those estimates will be consolidated into the total project budget at the end. This approach is suitable for the Smart Retail Inventory and Demand Prediction System because the project contains several distinct cost areas, and those include system design, AI/ML development, integration work, cloud infrastructure, testing, project management, training, change management, and also the contingency reserve that has been set aside.

The cost estimates are based on the approved project scope and also on those budget categories that were defined in the Project Charter. The key assumptions used in the estimation process are that the project will run from 26 February 2026 to 23 June 2026, the organization will keep using its existing ERP, POS, and WMS systems, the project team will not replace those existing legacy platforms, and the system will be hosted on an enterprise cloud platform like AWS or Azure. It is also assumed that the existing retail data from 2023 onwards is structurally usable for AI/ML model training, and that the organization's internal IT Department along with external vendors will provide the required technical support during integration and testing when it is needed.

Several constraints have been considered in the cost estimation as well. One thing is the total project budget is fixed at AUD 500,000, so all project costs must stay within that limit. Another thing is the

cloud infrastructure and hosting budget has been capped at AUD 80,000, and that cap requires active monitoring of GPU compute costs, storage costs, API hosting costs, and backup costs. A third aspect is the implementation timeline is pretty short, so cost planning has to make sure sufficient resources are available during those most intensive phases of development, integration, and testing. A fourth constraint is any unplanned scope expansion, any additional model tuning work, any vendor delay, or any API integration complexity that comes up may end up creating some additional cost pressure on the budget.

The Project Manager, the Finance Department, the Technical Lead, the QA and Change Management Lead, and also the relevant vendor representatives will review all of the cost estimates before they get approved. During the execution phase, cost control will be supported through cost baseline management, variance analysis, Earned Value Management, cost performance monitoring, contingency reserve management, and those formal change control procedures we have put in place.

Cost Breakdown

System Design & Architecture

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Technical Lead / System Architect	1	180	150	27,000
Business Analyst	1	150	100	15,000
Architecture design tools and documentation	1	-	8,000	8,000
Architecture review workshops and vendor consultation	1	-	10,000	10,000
Total				60,000

AI/ML Model Development

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
ML Engineers	2	300	130	78,000
Data Analytics Support	1	160	100	16,000
GPU training and experimentation environment	1	-	16,000	16,000

Model validation and technical documentation	1	-	10,000	10,000
Total				120,000

System Integration & Development

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Software Developers	2	300	110	66,000
API integration tools and middleware services	1	-	12,000	12,000
Automated replenishment engine development support	1	-	12,000	12,000
Integration documentation and code review	1	-	10,000	10,000
Total				100,000

Cloud Infrastructure & Hosting

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Cloud Engineer	1	180	100	18,000
Cloud hosting, compute, and storage services	1	-	50,000	50,000
Backup, recovery, and monitoring services	1	-	12,000	12,000
Total				80,000

Testing & Quality Assurance

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
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Quality Assurance / Testing Team	2	180	80	28,800
UAT coordination support	1	80	90	7,200
Testing tools and test environment setup	1	-	8,000	8,000
Defect tracking and quality reporting	1	-	6,000	6,000
Total				50,000

Project Management, Training & Change Management

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Project Manager	1	220	120	26,400
Change Management / Training Lead	1	120	70	8,400
Training materials, Quick Reference Guides, and LMS resources	1	-	5,200	5,200
Total				40,000

Contingency Reserve

Detail	Unit	Hours	Cost (AUD)	Total (AUD)
Contingency reserve for technical, integration, cloud, testing, and deployment risks	-	-	50,000	50,000
Total				50,000

Cost Baselines

The project cost baseline represents the approved time-phased budget across the project lifecycle. It will be used to monitor planned expenditure against actual expenditure and to support cost variance analysis. Since the project runs for approximately four months, the baseline is divided into four monthly periods.

WBS / Cost Area	Phase 1 Initiation & Planning	Phase 2 Design & Development	Phase 3 Integration & Testing	Phase 4 Deployment & Closure	Total (AUD)
System Design & Architecture	30,000	25,000	5,000	-	60,000
AI/ML Model Development	-	90,000	30,000	-	120,000
System Integration & Development	-	40,000	50,000	10,000	100,000
Cloud Infrastructure & Hosting	10,000	30,000	30,000	10,000	80,000
Testing & Quality Assurance	-	-	40,000	10,000	50,000
Project Management, Training & Change Management	10,000	10,000	5,000	15,000	40,000
Contingency Reserve	10,000	15,000	15,000	10,000	50,000
Total	50,000	210,000	175,000	65,000	500,000

The cost baseline is aligned with the project schedule. Month 1 focuses on initiation, planning, system design, early cloud setup, and project management. Month 2 has the highest planned expenditure because it includes major AI/ML development, system design completion, cloud infrastructure usage, and early system integration work. Month 3 includes continued integration, model validation, testing, and quality assurance activities. Month 4 covers final testing, deployment, training, go-live support, closure activities, and remaining contingency coverage.

The total cost baseline is **AUD 500,000**, which is equal to the total approved budget and the total cost breakdown. This ensures consistency between the Project Charter, cost breakdown, and cost baseline.

Funding requirements and sources

The Smart Retail Inventory and Demand Prediction System will be funded through the organization's internal digital transformation budget. This funding source is appropriate because the project is directly aligned with the organization's strategic objectives, and those objectives are improving operational efficiency, increasing forecast accuracy, reducing out-of-stock incidents, reducing excess inventory holding costs, and supporting long-term digital transformation across the company.

Funding will be released in phases according to how the project milestones get achieved. The first funding release will support project initiation, planning activities, system design work, and the early cloud setup. The second release will support AI/ML model development, architecture completion, and the initial integration work with existing systems. The third release will support system integration, cloud operation, testing, and quality assurance activities. The final release will support UAT, training, go-live, the hypercare period, and project closure tasks.

Vendor-related expenditure, and that includes cloud hosting costs, API tools, software services, and external support, will be managed through milestone-based payment arrangements wherever that is possible. This approach makes sure payment gets linked to deliverables that have been completed, and it reduces the risk of paying for work that is incomplete or unacceptable. The Finance Department will review project spending on a monthly basis, while the Project Manager will monitor expenditure on a weekly basis against that approved cost baseline that has been established.

The contingency reserve of AUD 50,000 will be used only for approved project risks that actually materialize, and examples of those would be unexpected integration issues, additional cloud scaling requirements that come up, testing defects, vendor delays, or deployment support needs. Any use of contingency funding has to be documented properly and approved according to that project governance process we have set up. Any cost increase that goes beyond the approved AUD 500,000 budget must be submitted as a formal change request and then approved by the Project Sponsor before any additional spending can happen.

Justification, Tools, Time, and Resource Estimation

Strategic justification of the IT project

Why this project?

The smart retail inventory and demand prediction system can help address several challenges that retail organizations tend to face out there. The company currently relies on a traditional, rule-based kind of inventory management system, and this system runs across different ERP platforms, POS platforms, and warehouse management systems. This fragmented infrastructure can lead to poor inventory outcomes, and what that means is inaccurate demand forecasting, delayed decisions, and inefficient stock management across all of the various locations.

So, the main proposed IT initiative is aiming to help solve these challenges by using an AI-driven inventory forecasting and demand prediction system. By using those machine learning models like LSTM and XGBoost, the system can analyze the historical sales data, customer purchasing patterns, and current market trends. Then, by using the real-time inventory information that comes in, the system can generate an accurate demand forecast. This will allow the organization to make proactive inventory decisions instead of relying on a manual process that takes more time.

The system will also aim to integrate with the company's existing ERP and POS platforms through secure APIs, and the purpose of that is to provide real-time inventory visibility and tracking. Automated purchase order generation and dynamic safety stock calculation will help minimize human error, reduce stock outages, and improve those inventory turnover rates across the various retail locations that the company operates.

Currently, it is a very good time to implement this solution in the retail industry, and the reason for that is because the industry is becoming extremely competitive and increasingly data-driven. Customers now expect much faster product availability, accurate order fulfillment, and a very seamless shopping experience, whether they are shopping on digital platforms or physical ones. All of the other competitors out there are also adopting these AI-powered retail technologies, and they are doing that in order to improve their operational efficiency and strengthen their supply chain performance. Without any digital transformation happening, the organization runs the risk of falling behind and losing market share to those competitors.

So, the project strongly aligns with the strategic goals of the company, and those goals are to improve efficiency, reduce inventory holding costs, and enhance customer satisfaction by supporting long-term digital transformation initiatives. By improving forecast accuracies, the system is expected to reduce those stock fallout incidents, minimize the amount of excessive inventory being held, and improve decision-making across the board, which basically leads to an overall improvement in profitability.

Additionally, the project will also support scalability and future opportunities for the business's growth by providing a cloud-based infrastructure, one that is capable of supporting various retail locations with larger data sets and with future AI improvements that might come along. Through these automations, data analytics, and real-time inventory management, the smart retail inventory and demand prediction system will also create long-term business value and strengthen the competitive position of the organization within this market sector.

Alignment with Business Strategy

The smart retail inventory and demand prediction system aligns with the organization's objectives, and those objectives are to improve operational efficiency, improve customer satisfaction, and also reduce the operational costs that the company has to bear. Not only that, but the organization also aims to support a long-term digital transformation over time. As the retail industry is becoming increasingly competitive and also very data-driven, every organization out there has to adopt these advanced technological changes so that they can remain in the competition and also meet those evolving requirements that each customer has. The proposed system helps to support this transition by enabling predictive decision-making, automated inventory management, and real-time operational visibility across all of those various retail locations.

One of the most important objectives that gets supported by this project is the improvement of operational efficiency. The current inventory management approach basically relies pretty heavily on manual decision-making and on those divided systems that don't talk to each other, and what this often results in is stockouts, overstocking, or a delayed replenishment process. So by implementing that AI-driven forecasting model, the organization can improve the accuracy of this demand prediction system and help avoid these kinds of problems from happening in the future. This will also reduce the manual workload that staff have to deal with, minimize those operational inefficiencies, and improve the supply chain responsiveness as well.

The project also helps to support customer satisfaction and thus builds trust toward the company from its customers. Accurate inventory forecasting and real-time inventory visibility help to make sure the product remains available during those periods when there is a high level of customer demand. So reducing those stockout incidents will improve the customer experience and their satisfaction, and that will build the customer's trust and further strengthen the brand's reputation. In a modern retail environment, product availability and efficient order fulfillment are very critical factors when it comes to influencing a customer and having a competitive advantage over other retailers.

The project supports the organization's roadmap to digital transformation by introducing a cloud-based infrastructure, AI technologies, and also some advanced analytical capabilities. This digital transformation initiative will provide a scalable technology foundation, and that foundation will support future business growth, expansion into more retail locations, and further AI-driven innovations that might come later. The project delivers a very clear value proposition by improving the forecasting accuracy, reducing those operational inefficiencies, and also enabling proactive inventory management. If you look at it from a financial perspective, the organization is expected to achieve a strong return on investment through reduced stock losses, improved operational productivity, lower inventory holding costs, and increased sales opportunities that come from better product availability.

The expected project outcomes include:

- Reduction in stockout incidents
- Improved inventory turnover
- Increased forecast accuracy
- Lower excess inventory costs
- Enhanced operational efficiency
- Improved customer satisfaction

- Better data-driven decision making

Chosen Systems/Tools/Technologies

System/Technology options considered

Tool/System	Description	Pros	Cons
LSTM (Long Short-Term Memory)	It is a deep learning model that can be used for time-series forecasting and demand prediction by analyzing the historical data of customer behaviour and sale trends.	• Predicts demand trends accurately. • Reduces stockouts and overstocks. • Learns sales patterns effectively. • Supports better inventory planning decisions.	• Requires large amounts of historical data • Needs high computing power • Complex to develop and maintain • Can increase implementation costs
XGBoost	It is a machine learning algorithm which is used for predictive analytics and retail forecasting. It helps to identify relationships between sales trends and inventory movement.	• Fast and efficient for analysing retail datasets • Improves forecasting accuracy • Scalable for large business operations • Works well with structured inventory data	• Requires technical expertise for setup • Parameter tuning can be time-consuming • Results may be difficult for non-technical users to understand
Cloud Hosting Services	It provides a scalable cloud infrastructure and storage required for real-time inventory monitoring and forecasting operations.	• Reduces need for physical hardware • Supports remote access and scalability • Improves system flexibility and availability • Easier to support future business growth	• Ongoing subscription and hosting costs • Dependent on stable internet connection • Third-party provider downtime may affect operations
API Integration Middleware	It is used to connect ERP, POS, and Warehouse Management Systems with the forecasting	• Enables real-time data sharing between systems • Reduces manual data entry errors	• Integration with older systems may be difficult

	platform with the help of APIs.	Improves operational efficiency	- Setup and testing can take time - Requires strong cybersecurity protection
Cybersecurity and access control system	It is security tool used to protect inventory, operational, and customer data through monitoring and authentication controls.	- Protects sensitive business and customer data - Reduces risk of cyberattacks and data breaches - Supports regulatory compliance - Improves system reliability and trust	- Adds additional implementation costs - Requires regular updates and monitoring - Security setup may increase deployment complexity
Analytics and Dashboard Tools	It is used to visualise forecasting reports, inventory levels, and operational performance.	- Easy for managers to monitor inventory in real time - Improves decision-making using visual reports - User-friendly and accessible - Helps track operational performance quickly	- May require staff training - Dashboard customisation can increase development time - Ongoing maintenance and updates may be needed

Final Selection

After evaluating the multiple technologies and options for the system, the organization has selected a combination of LSTM (which is the long short-term memory model), XGBoost, a cloud-based infrastructure, and some API integration middleware for that smart retail inventory and demand prediction system. These technologies have been chosen because they provide very strong forecasting accuracy, good scalability, and compatibility of the system with the long-term operational values for the organization's retail operations.

LSTM has been selected as one of the primary forecasting models because we know it is highly effective for analyzing the time series patterns of sales data and for identifying those long-term demand patterns that exist in the data. Since retail demand tends to be heavily influenced by seasonal trends, customer purchasing behavior, and market fluctuations, LSTM will provide better forecasting accuracy when compared against traditional statistical forecasting methods. So, this can allow the organization to make proactive inventory decisions and reduce those stock shortage situations or overstock situations. Now, XGBoost has been selected to complement that LSTM forecasting model because of its speed, its efficiency, and its ability to process the structured retail data as required by the system.

XGBoost can help improve the predictive performance by identifying the complex relationships between demand fluctuations, sales variables, and also inventory levels. Combining these two models together can provide a hybrid forecasting approach, and that approach can improve the accuracy and operational reliability we get from the system. A cloud-based infrastructure has been selected because it offers good scalability and cost efficiency for the organization. Cloud hosting can also allow the system to support multiple retail locations, large data sets, and real-time inventory processing without requiring any major on-premises hardware investments from the company. The environment also supports business expansion and remote system access for operational staff and management teams. When you compare it to a traditional on-premise infrastructure, the cloud reduces maintenance costs and improves availability along with disaster recovery capabilities.

Now, API integration middleware has been selected to handle the communication between the smart retail inventory and demand prediction system and the organization's existing ERP, POS, and warehouse management systems. Since the organization currently operates on multiple disconnected systems, this middleware is very essential for enabling real-time data exchange, an automated replenishment process, and accurate inventory visibility across all of those retail locations. The middleware solution also minimizes manual data entry and reduces operational inefficiencies that have been happening.

The technologies that have been selected provide a strong cost-benefit advantage for the organization. Although the initial implementation costs are relatively high because of the AI development work, the integration activities, and the cloud infrastructure setup, the long-term operational savings are expected to outweigh those upfront expenses. Improved forecasting accuracy will help reduce excessive inventory holding costs, minimize those stockout incidents, improve the inventory turnover rate, and increase customer satisfaction. Because of this, the project is expected to generate a positive return on investment over a reasonable time period.

The selected systems provide a strong compatibility advantage and usability advantage as well. Cloud-based dashboards and automated forecasting tools improve usability for store managers, the procurement team, and the inventory controllers by providing real-time inventory tracking and data-driven insights they can act on. The technologies are also very compatible with the existing ERP, POS, and WMS systems, and that compatibility reduces the need for a complete replacement of the existing infrastructure. Overall, the selected technologies can provide a scalable, efficient, and future-ready solution to support the organization's digital transformation strategies and its long-term operational goals.

Time, Cost and Resource Estimation

Project Duration

Phase	Description	Start Date	End Date	Total Duration	Buffer days
Project Initiation	The main objective is Project Planning, requirement Identification and Stakeholder approval	26 February, 2026	04 Mar, 2026	1 Week	2 Days
Requirement Gathering	All the requirements are gathered which involves operational, functional and	5 March, 2026	25 March, 2026	3 Week	3 Days

	business type requirements.				
System Design & AI Development	This phase involves the designing of predicted architecture and development of the LSTM and XG Boost Model.	26 March, 2026	29 April, 2026	5 Week	5 Days
System Integration	Involves the integration of ERP, POS, and WMS systems using APIs.	30 April, 2026	13 May, 2026	2 Week	3 Days
Testing & Quality Assurance	In this phase different testings are done which involves functional, system and user acceptance testing.	14 May, 2026	03 June, 2026	3 Week	4 Days
Training & Deployment	Involves staff training, onboarding and the deployment of the project.	04 June, 2026	23 June, 2026	3 Week	2 Days

Budget Breakdown

Category	Estimated Cost (AUD)	Notes
System Design & Architecture	60,000	Technical planning, forecasting architecture, integration design, and dashboard design.
AI/ML Model Development	120,000	Development of LSTM and XGBoost models for demand prediction.
System Integration & Development	100,000	Integration of ERP, POS, and Warehouse Management Systems, including API middleware and replenishment logic.
Cloud Infrastructure & Hosting	80,000	Cloud servers, storage, compute resources, hosting, backup, and monitoring.
Testing & Quality Assurance	50,000	Functional testing, integration testing, system testing, UAT, and defect management.

Project Management, Training & Change Management	40,000	Project coordination, stakeholder meetings, governance reporting, staff training, onboarding, and change support.
Contingency Reserve	50,000	Reserved for unexpected technical, operational, integration, or deployment issues.
Total Budget	500,000	Approved project budget.

Resource Planning

Phase	Resource	Allocation	Notes
Project Initiation	Project Manager	Full-Time	Oversees project planning, scheduling, communication, and governance activities.
Requirement Gathering	Business Analyst	Full-Time	Collects operational and technical requirements from stakeholders and retail teams.
System Design	Technical Lead and Developer	Full-Time	Designs forecasting architecture, dashboards, APIs, and database structure.
AI Model Development	ML Engineers	Full-Time	Develops and trains LSTM and XGBoost forecasting models for demand prediction.
System Integration	Software Developers	Full-Time	Integrates ERP, POS, and Warehouse Management Systems (WMS) using APIs.
Cloud Infrastructure Setup	Cloud Engineer	Part-Time	Configures cloud hosting, storage systems, backup, and network infrastructure
Testing and Quality Assurance	QA Test Team	Part-Time	Conducts functional testing, integration testing, and user acceptance testing.
Cybersecurity Implementation	Cybersecurity Specialist	Part-Time	Implements security monitoring, access

			control, and data protection measures.
Training and Deployment	Change Management Team	Full-Time	Provides staff training, onboarding support, and deployment assistance.
Post- Go-Live Support	Help Desk Team	Part-Time	Resolves technical issues and provides operational support after system deployment

Risk Management Plan

Introduction

There is an essential role of risk management to the planning and delivery of the Smart Retail Inventory and Demand Prediction System, and a bunch of uncertainty comes with this project in several different areas; one thing is the machine learning performance, another thing is the data quality, a third aspect could be cloud reliability, plus cybersecurity, vendor coordination, and also the integration work with those existing systems like ERP, POS, and WMS. If the project team doesn't address these kinds of issues early on, there could be delays, an increase of costs to the project, a reduction of system quality, or even a weakening of user acceptance could happen because of it. For this reason, you would want to treat risk management as an activity that goes on throughout the whole project, not just as a planning task that gets completed at the very beginning, of course.

With this Risk Management Plan, the purpose of it is to set up a process that's practical for managing all of the project's risks through the project's whole lifecycle, and the plan goes on to explain how the project team would be able to identify potential problems, assess the likely impact of them, and also decide on response actions that are suitable to take. On one hand, it clarifies the way responsibility for each risk gets assigned; on the other hand, there is a clarification of how the effectiveness from mitigation actions is going to be reviewed over a period of time, and this part is particularly important because the project really depends on both the technical delivery side of things and the organizational adoption side of things too. To give you an example, poor data quality could reduce the forecast accuracy, delays coming from different vendors might end up affecting the integration work, and a limited amount of user acceptance would probably reduce the value of that system after you have deployed it, of course.

Another thing the plan recognizes is that there could be opportunities created by uncertainty sometimes, and there is improved monitoring which may be able to strengthen the system's reliability; at the same time, automated generation of purchase orders could reduce the amount of manual procurement effort more quickly than anyone would have expected. Where these kinds of opportunities are able to support the project's objectives, they should be recorded and managed by the team in a way that's pretty similar to how you would handle other project risks, of course. So overall, what this Risk Management Plan does for the project team is to help with controlling uncertainty, responding to issues in a timely enough way, and also improving the likelihood of a successful deployment and adoption by everyone involved.

Risk Management Approach

For this project, there is going to be an ongoing kind of approach to risk management, and it is based on those risk management principles from PMBOK, of course; risk management will not be treated as a task that gets completed only during the planning phase, but instead it will be built right into the whole lifecycle of the project. One thing is it begins at the initiation stage, another thing is it continues through planning, then development, then testing, then deployment, and also through that post-go-live period they call hypercare.

Risk identification will start during the project planning phase, and then it will keep on going as the project moves along; the very first set of risks is going to be developed by reviewing some key documents from the project, and those documents include the Project Charter, the Scope Management Plan, the Schedule Management Plan, the Stakeholder Management Plan, plus the Project Governance and Roles document too. On one hand, the project team will also identify risks by running stakeholder workshops and interviews, looking at previous lessons learned from other IT projects that were similar, and talking

with both the technical teams and the business teams for their input. On the other hand, for this particular project, there will be a special kind of attention given to those risks that are linked to AI/ML model performance, data quality, uncertainty around cloud costs, integration with legacy systems, delays from vendors, cybersecurity compliance, and also resistance from users, of course.

When it comes to risk analysis, the project will mainly use a qualitative kind of assessment, and this will be supported by a limited amount of quantitative analysis but only in those cases where the impact can be estimated with some reasonable confidence. Each risk is going to be assessed by means of a probability-impact matrix, and the purpose of that is to figure out what kind of effect the risk would likely have on the project's objectives; based on this assessment, the risks will then be rated as Low, Medium, High, or Extreme. There is a quantitative analysis that will only be used for those major cost-related or schedule-related risks though, for example a cloud resource overrun, a delay from a vendor, or an API integration failure, because those are the kinds of situations where the likely time impact or financial impact could be estimated with enough confidence, of course.

For risk response planning, the team will define actions that are suitable for the major risks; one thing is negative risks could be avoided, reduced, transferred, accepted, or even escalated depending on how severe they are and also depending on the level of control that the project team has available to them. Another thing is positive risks are going to be considered as well, but only where they would be able to improve the project's outcomes, like a faster procurement automation process or a better kind of reliability from the system. Each significant risk will end up being assigned to a risk owner, and that owner will be responsible for monitoring early warning signs, coordinating the response actions, and also providing updates to the Project Manager from time to time. There is a risk register that will be reviewed on a weekly basis by the project team, plus it will be reviewed on a monthly basis by the Steering Committee, and the reason for doing that is to make sure the risk information stays current and useful, of course.

Risk Identification

There is a risk identification process that will be conducted as an iterative kind of activity throughout the whole lifecycle of the project, of course; the very first cycle of risk identification will happen during project planning, and the inputs that get used for it are the approved project scope, the approved schedule, the stakeholder analysis, and also the governance structure that has been set up. Because the system ends up depending on quite a few technical components and also organizational components, the project team is going to use a Risk Breakdown Structure, and the reason for that is to make sure that risks get considered in a systematic way across all of these categories: technology, data, people, financial, vendor, compliance, and process.

Several different techniques will be used by the team to capture the risks; one thing is there will be brainstorming workshops held with the Project Manager, the Technical Lead, the Business Analyst, the QA and Change Management Lead, the IT Department, and also those Inventory and Procurement representatives. These workshops are going to be pretty useful for identifying broad delivery risks, and examples of that would include integration delays, user resistance, and requirements that are unclear. Another thing is structured interviews will be conducted with subject matter experts, and that group includes data analysts, store operations representatives, finance staff, and vendor contacts, and the purpose of these interviews is to uncover those risks that might not be visible to the core project team. A third aspect is a SWOT analysis will also be applied, and the purpose of that is to identify both threats and opportunities; to give you an example of a threat, there is the risk of poor model accuracy, and an example of an opportunity would be the chance to improve operational decision-making by means of predictive analytics.

The risk register is going to be treated as a living kind of document, of course; new risks could be added during sprint reviews, status meetings, quality reviews, UAT sessions, procurement reviews, or even change control discussions. Any risk that has a potential impact on scope, schedule, budget, system quality, compliance, or adoption will be recorded by the team, then assigned an owner to it, then analyzed, and then monitored over time as well.

Risk Analysis

Risk analysis is going to be used by the project team to prioritize those risks and also to figure out what level of management attention is required for each one of them; the project will primarily use a qualitative kind of risk analysis, and the reason for that is because a lot of the risks that come with AI-enabled IT projects, like user resistance, model uncertainty, and integration complexity, cannot really be estimated in a precise way at the very beginning of the project, of course. Each risk that gets identified will be assessed according to two dimensions: one thing is its likelihood of occurrence, and another thing is its impact on the project's objectives.

The likelihood is going to be rated as Low, Medium, or High; the impact will also be rated as Low, Medium, or High, and this rating is based on the potential effect that the risk could have on cost, schedule, scope, quality, security, compliance, or even stakeholder acceptance. There is a combination of likelihood and impact that will end up determining what the overall severity level is going to be; any risks that have both a high likelihood and a high impact will be classified as Extreme, and those ones will get escalated up to the Steering Committee. For those risks with a high severity level, they will require active mitigation and also a weekly kind of monitoring; for those Medium risks, they will require an assigned owner and then a periodic review of them, and for the Low risks, they will simply be recorded and then monitored by means of the normal project reporting that already happens.

Where it is possible to do so, there will be a quantitative analysis applied to those risks that have measurable cost consequences or schedule consequences; to give you an example, a cloud GPU cost overrun could be estimated by using the expected usage variance, and a vendor delay could be assessed by estimating the number of affected work packages that sit on the critical path. This limited kind of quantitative analysis will be able to support decisions about contingency reserves, schedule buffers, procurement alternatives, and also escalation thresholds, of course. The output that comes from the risk analysis is going to be an updated risk register, and that register will rank all of the risks by their priority and also provide input that goes into the response planning work.

Risk Response Planning

Risk response planning is what defines how the project team will be able to address those prioritized risks before they end up threatening the success of the project, of course. For the negative risks, the project is going to use five different response strategies: one thing is avoid, another thing is mitigate, a third aspect is transfer, a fourth one is accept, and the fifth one is escalate. There is an avoidance strategy that will be used in those cases where the project can remove a risk by changing the scope or the approach, for example by excluding those non-essential dashboard features from the very first release. Mitigation is going to be the most common strategy that gets used, and it will be applied to reduce either the likelihood of a risk or the impact from it, such as conducting some early data profiling activities to reduce any delays in model training. A transfer strategy will be applied in those situations where a risk can be shifted over contractually to the vendors, and this would be done by means of service-level agreements, warranties, or support obligations that the vendor has to follow. There is an acceptance strategy that will be used for those low-priority risks where the cost of doing a response

would actually exceed the likely impact from the risk. Then escalation will be used for those cases where a risk ends up exceeding the Project Manager's authority, and examples of that would be a major budget increase, a regulatory change, or an enterprise-wide kind of cybersecurity exposure.

Now for the positive risks, the project will use a different set of strategies, and those are exploitation, enhancement, sharing, acceptance, and escalation as well. To give you an example, if early model testing is able to achieve a forecast accuracy that is higher than what was expected, the team could exploit that opportunity by accelerating the dashboard demonstration activities and also by doing more stakeholder engagement work. Another example would be if those AI-powered monitoring tools are able to reduce the post-deployment support risk, then the project could enhance this particular opportunity by integrating the monitoring earlier on into the deployment plan, of course.

Each one of the major risks is going to have a risk owner, and that owner will be responsible for monitoring the triggers and also for executing the response actions that have been planned. There is a Project Manager who will maintain the overall risk register, review the status of each risk during the weekly project meetings, and also escalate those High or Extreme risks up to the Steering Committee when needed. All of the response actions will be reviewed for their cost-effectiveness, their feasibility, and also for how well they align with the project's scope baselines, schedule baselines, and quality baselines.

Risk Register

Risk No.	Risk Name	Risk Description	Risk Owner	Mitigation Plan	Impact Level	Likelihood of Occurrence	Severity Level	Contingency Plan
R1	Poor Data Quality	Historical POS, WMS, and ERP data may contain missing values, duplication, inconsistent SKU codes, or inaccurate inventory records, reducing ML forecast accuracy.	Data Analytics Team / Technical Lead	Conduct early data profiling, cleansing, validation rules, and sample reconciliation before model training.	High	High	Extreme	Extend data preparation window, use fallback rule-based forecasting for affected SKUs, and escalate data ownership issues to the Project Sponsor.



R2	ML Model Accuracy Below Target	LSTM/XGBoost models may fail to achieve the target forecast accuracy of $\geq 85\%$ due to insufficient data quality, seasonality, sparse SKU history, or cold-start items.	Technical Lead / ML Engineer	Use baseline models, feature engineering, cross-validation, error analysis, and cold-start heuristics.	High	Medium	High	Reduce initial model scope to high-volume SKUs, deploy hybrid rule-based + ML forecasting, and continue model improvement during hypercare.
R3	Legacy ERP/POS/WMS Integration Delay	API development and integration with legacy systems may take longer than expected due to undocumented interfaces, vendor constraints, or technical incompatibility.	Technical Lead / IT Department	Conduct early interface discovery, confirm API documentation, create integration prototypes, and hold regular vendor technical reviews.	High	High	Extreme	Use temporary batch-file integration, re-sequence non-dependent tasks, and escalate vendor issues through the Steering Committee.
R4	Vendor Delivery Delay	Cloud provider, ERP vendor, ML solution	Project Manager / Vendor Manager	Define clear SLAs, milestone-based	High	Medium	High	Activate backup vendor support, reallocate

		vendor, or integration consultants may delay required deliverables or technical support.		deliverables, vendor reporting cadence, and penalty clauses.				internal resources, or escalate contract non-performance to procurement governance.
R5	Cloud Cost Overrun	GPU training, storage, data transfer, monitoring tools, and cloud-based MLOps services may exceed budget due to consumption-based pricing.	Project Manager / Finance Department	Set cloud budgets, alerts, usage dashboards, reserved capacity where suitable, and approval thresholds for high-cost workloads.	High	Medium	High	Use contingency reserve, reduce training frequency, optimise compute usage, or move non-critical workloads to lower-cost resources.

Quality Management Plan

Introduction

Quality in this project refers to the degree to which that Smart Retail Inventory and Demand Prediction System is able to meet the documented requirements, meet the stakeholder expectations, and also meet its intended business purpose, of course. In IT projects, quality is not just limited to whether the software functions correctly or not; it also includes a bunch of other things like reliability, performance, usability, maintainability, and security. This aligns with what was said in the Week 05 lecture, where the definition was that quality concerns whether a product or service meets the requirements, meets the expectations, and is fit for whatever its intended purpose is supposed to be.

For this particular project, quality management is absolutely essential, and the reason for that is because the system is going to directly affect inventory forecasting, replenishment decisions, supplier coordination, and also those daily store operations that happen all the time. If there is poor quality coming out of the project, that could end up resulting in inaccurate demand forecasts, data synchronization errors, operational disruption, or even a low level of user trust from the people who have to use it. Because of this, the Quality Management Plan that has been put together covers both product quality and also process quality. On one hand, product quality focuses on whether the system deliverables are able to meet the functional standards, the performance standards, the security standards, and the usability standards; on the other hand, process quality focuses on whether the development activities, testing activities, reporting activities, and approval activities are performed in a way that is consistent and effective. There is a plan that includes both quality assurance, which is process-oriented and preventive by nature, and quality control, which is product-oriented and verification-based.

Objective

The objective with this Quality Management Plan is to define how quality is going to be planned, managed, measured, and controlled throughout that whole Smart Retail Inventory and Demand Prediction System project, of course. One thing is it establishes some clear quality standards and also measurable criteria for the major deliverables, and those deliverables include the AI forecasting engine, the API integration middleware, the operational dashboard, the automated replenishment engine, and all of the project documentation that gets produced. Another thing is it makes sure that quality gets built into the project right from the very beginning, and this is done by means of preventive activities like process audits, peer reviews, coding standards, requirements traceability, and also a risk-based kind of defect prevention. A third aspect is the plan defines some quality control activities, and those activities are there to verify whether the completed deliverables are able to meet the agreed requirements before anyone accepts them; examples of these activities would be unit testing, integration testing, system testing, performance testing, security checks, and also User Acceptance Testing. This follows that Week 05 lecture distinction, where there is a difference between managing quality as a preventive process on one hand and controlling quality as a verification process on the other hand. Then finally, the plan also aims to support stakeholder satisfaction by making sure that the system which gets delivered ends up being reliable, secure, usable, and properly aligned with the organization's business objectives.

Scope

The scope of quality management covers both the outputs that get produced by the project and also the processes that are used to produce those outputs, of course. In terms of the product deliverables, quality is going to be measured across quite a few different things: the AI demand forecasting module, the data ingestion and ETL pipeline, the API integration middleware, the automated replenishment engine, the web-based operational dashboard, the MLOps retraining pipeline, the security controls, and also the user-facing documentation that people will have to read. There is an AI forecasting module that will be assessed against forecast accuracy criteria and model validation criteria; the integration middleware will be assessed against data integrity, latency, availability, and also whether it achieves successful synchronization with those ERP, POS, and WMS systems. Then the dashboard and the replenishment engine will both be assessed against functional correctness, usability, exception handling, and the accuracy of the business rules that have been put in place.

The quality scope also includes a bunch of project management deliverables as well, and examples of those would be the Project Management Plan, the Requirements Traceability Matrix, the test plans, the defect logs, the UAT sign-off records, the training materials, and also the post-implementation review documentation. These documents are pretty important, and the reason for that is because they are able to provide evidence that the project work has been properly planned, properly tested, properly reviewed, and then formally accepted by the right people, of course.

Now in terms of process quality, the plan ends up covering the development practices, the testing procedures, the change control process, the defect management process, the reporting activities, and also the stakeholder review activities. There is a Week 05 lecture that really emphasizes how project quality management includes both the quality of the deliverables on one hand and the quality of the project processes on the other hand. Because of this, the plan is not only going to check whether the final system actually works or not; it will also check whether the team follows the agreed standards, conducts the appropriate reviews, maintains traceability throughout the work, and resolves all of those defects before the go-live date arrives.

Quality Standards and Criteria

The quality standards for this project are based mainly on those project quality management principles that were covered in the Week 05 lecture, of course. The first applicable standard is ISO 9001, and what that one does is it provides a general kind of quality management system approach, and the focus of it is on continuous improvement, customer satisfaction, and having processes that are properly documented. For this project, those ISO 9001 principles will be reflected by means of formal requirements documentation, quality metrics, review checkpoints, and also some corrective action processes that the team can use.

Now the second relevant standard is ISO/IEC 27001, and the reason that one matters is because security itself is a core dimension of IT quality in general. Since the system is going to process a bunch of retail operational data and also integrate with those ERP, POS, and WMS platforms, the quality of it has to include the confidentiality of data, the integrity of data, and also the availability of data. There are some security-related criteria that will be included, and examples of those would be role-based access control, encryption both in transit and also at rest, secure API communication, and also a guarantee of no critical data leakage or corruption happening along the way.

CMMI is also relevant to this project, and that is because it focuses on assessing and improving software development processes in a pretty systematic way. For this project, the team should be aiming for a managed and defined kind of process, where the requirements, the testing activities, the defect resolution work, and the deployment activities are all documented and also repeatable from one iteration to the next.

There are also ITIL principles that become applicable after the deployment happens, and that is because the system will end up becoming an operational IT service once it is live; ITIL is able to support service quality by means of service level agreements, continuous improvement, and also alignment with whatever the business needs happen to be.

The measurable quality criteria for this project are as follows, of course: one thing is the forecasting module should be able to achieve at least eighty-five percent SKU-level forecast accuracy; another thing is API data synchronization should have zero critical data loss or corruption happening to it; a third aspect is system uptime should reach at least ninety-nine point five percent after the deployment has been done; API response latency should remain below whatever the agreed threshold is during those peak operating hours; all Severity 1 defects and Severity 2 defects must be resolved before the go-live date arrives; and also UAT sign-off must be obtained from the key business users who will be using the system. All of these criteria align really well with the project success criteria that were stated earlier in the Project Charter.

Quality Assurance and Control Activities

Quality assurance activities are going to focus on preventing defects from happening in the first place and also making sure that the project processes get followed in the correct way, of course. There is a QA Lead who will be responsible for maintaining the test strategy, defining the quality metrics that the team will use, conducting process audits from time to time, and also ensuring that all of those development practices and testing practices end up complying with the approved Quality Management Plan. Those process audits are there to check a bunch of different things: one thing is whether the developers actually follow the coding standards that have been set up; another thing is whether the user stories include proper acceptance criteria; a third aspect is whether the test cases are properly linked to the requirements; and also whether the CI/CD logs and the defect records are maintained in a proper way. This really reflects that Week 05 lecture explanation, where managing quality translates the Quality Management Plan into quality activities that can actually be executed, and the focus of it is on preventing defects rather than just detecting them after they have already happened.

Quality assurance is also going to include a few other activities like peer code reviews, architecture review sessions, sprint retrospectives, requirements traceability checks, and also some security design reviews. These activities are intended to help identify any process weaknesses early on, reduce the amount of technical debt that builds up, and also improve the collaboration between the development team, the QA team, the business team, and the operations team. To give you an example, there is a Requirements Traceability Matrix that will be used to make sure each approved requirement ends up being linked to a design component, linked to a test case, and also linked to an acceptance result at the end of the process.

Now quality control activities are going to focus more on inspecting and validating the deliverables that have already been completed by the team. The testing team will perform a bunch of different tests: one thing is unit testing for the individual software modules; another thing is integration testing for those interfaces between the AI engine and those ERP, POS, and WMS systems; a third aspect is system testing for the complete solution as a whole; there will also be performance testing to check API latency and system responsiveness; security testing will be done for access control and data protection; and finally there will be User Acceptance Testing done together with the Store Managers and the Procurement Team. The Week 05 lecture identifies unit testing, integration testing, system testing, and UAT as the key types of testing that matter for quality control purposes. Any defects that get found will be logged into the system, prioritized by their severity level, assigned to owners who are responsible for fixing them, retested after the correction has been made, and then reviewed one more time before anyone accepts

them. Deliverables will only be accepted when they are able to satisfy the defined quality criteria and also receive a formal sign-off from the right people.

Communication Management Plan

Objectives

Communication is necessary for making sure project information gets delivered to the right stakeholders, through the right channels, at the right time, and in the right format. For the Smart Retail Inventory and Demand Prediction System, effective communication is particularly important because the project involves quite a few internal and external stakeholders, and those include executive sponsors, IT teams, inventory and procurement users, store managers, cloud providers, ML vendors, ERP vendors, and also system integration consultants. All of these stakeholders have different levels of power, different levels of interest, and different information needs, so a single informal communication approach would not be enough to cover everything.

The purpose with this Communication Management Plan is to define how project information will be collected, distributed, stored, retrieved, and monitored throughout the project's whole lifecycle. This aligns with the project communication management process that was covered in the lecture slides, where communication planning is what identifies the stakeholder communication needs, the communication methods, the roles and responsibilities, the frequency of communications, and the feedback mechanisms that will be used. The plan also helps with managing and monitoring communications by making sure the information flow remains accurate, stays timely, and aligns with what the stakeholders expect to receive.

This plan will help reduce misunderstandings, duplicated work, delayed decisions, and also stakeholder resistance. It will also support project governance by creating clear reporting lines, clear escalation paths, and documentation standards for project status updates, risks, issues, changes, testing activities, deployment work, and post-go-live support activities.

Scope

The scope of this Communication Management Plan covers all formal project communications that are needed to support the planning, execution, monitoring, control, and closure of that Smart Retail Inventory and Demand Prediction System. It applies to communications between the project team, the executive sponsor, the steering committee, the business users, the technical teams, the vendors, and any other relevant stakeholders who have a role to play. The plan includes communication related to project status reporting, schedule updates, budget updates, risk and issue escalation, change requests, quality and testing updates, UAT coordination, training notices, vendor coordination, deployment readiness, and also the post-implementation review that happens after go-live.

This plan does not cover informal personal conversations, those internal communications that happen within a vendor's own team, unrelated organizational announcements, or any detailed technical documentation that gets managed under separate technical workstreams. To give an example, source code documentation, system architecture specifications, detailed API documentation, and user manuals may be communicated through this plan from time to time, but the actual technical content of them is managed separately by the relevant technical owners who are responsible for those items.

The Communication Management Plan supports the overall Project Management Plan by linking stakeholder engagement, governance, schedule control, risk management, quality management, and change management all together. In the project lifecycle, planning is what defines the roadmap, execution

is what implements the work and the communication plan, and then monitoring and controlling is what tracks the performance, the risks, the quality, the resources, and the communications. Because of this, the plan makes sure communication is not treated as some ad-hoc activity, but rather as a controlled management process that runs across the entire project lifecycle.

Stakeholder information requirements

The information requirements for each stakeholder are based on their level of interest, their project role, and their degree of influence over the project. The Project Sponsor and Steering Committee need high-level information about project progress, budget status, milestone achievement, major risks, unresolved issues, and those decisions that require executive approval. Their communication should focus on governance, strategic alignment, and exception-based reporting rather than getting into all the operational details.

The Project Manager needs detailed and continuous information from all of the workstreams, and that includes scope progress, schedule performance, cost status, resource issues, vendor updates, risks, defects, and stakeholder concerns. This information is needed so the Project Manager can coordinate delivery, maintain the project baseline, and escalate issues whenever it becomes necessary. The Technical Lead, ML Engineer, IT Department, and Data Analytics Team all need technical information like infrastructure readiness, data quality, model performance, integration dependencies, testing defects, and deployment constraints. Their communication has to be frequent because technical delays or integration issues can directly affect the project schedule and the quality of what gets delivered.

The Inventory and Procurement Team and the Store Managers are the primary business users for this system. They need information about requirements validation, workflow changes, UAT activities, training arrangements, go-live impacts, and support channels. Their communication should be clear and practical and focused on adoption, because poor user understanding could lead to resistance or inconsistent use of the system. Finance needs budget and cost variance information from the project, while external vendors need technical specifications, delivery expectations, issue logs, service levels, and formal change notices. Any regulatory or compliance stakeholders out there need milestone-based evidence that privacy obligations, security obligations, and compliance obligations are being properly addressed.

Communication Table

Stakeholder Name	Level of Interest	Description of interest	Document Name	Communication Medium	Frequency of communication
Project Sponsor	High	Requires visibility of project progress, budget status, major risks, milestone performance and strategic alignment.	Executive Status Report; Steering Committee Brief	Steering Committee meeting; email report	Monthly; immediate escalation for critical issues
Steering Committee	High	Requires governance-	Steering Committee	Formal meeting;	Monthly

		level updates and decisions on major scope, cost, schedule, risk or vendor issues.	Report; Decision Log	presentation; email summary	
Project Manager	High	Requires detailed information across all workstreams to coordinate delivery, manage baselines and control risks/issues.	Project Status Report; Risk Register; Issue Log; Change Log	Jira/Confluence; Teams; email; meetings	Daily/weekly depending on project phase
Business Analyst	High	Requires stakeholder feedback, requirements clarification, process change information and UAT inputs.	Requirements Register; RTM; UAT Feedback Log	Workshops; Teams; email	Weekly; more frequently during requirements and UAT
Technical Lead / ML Engineer	High	Requires technical requirements, model performance updates, integration dependencies, architecture decisions and defect information.	Technical Progress Report; Model Validation Report; Integration Log	Daily stand-up; Jira; technical meetings	Daily during development ; weekly otherwise
IT Department	High	Requires infrastructure, security, access control, cloud environment, deployment and operational support information.	Infrastructure Plan; Security Checklist; Deployment Plan	Technical workshops; Teams; email	Weekly; ad-hoc for incidents or deployment issues
Inventory & Procurement Team	High	Requires requirements validation, replenishment	Requirements Document; UAT Plan; Training Materials	Workshops; UAT sessions; email	Fortnightly; weekly during UAT and training

		workflow updates, UAT instructions and training information.			
Store Managers	High	Require information about operational impacts, go-live timing, training arrangements, support channels and dashboard use.	Go-Live Notice; Training Schedule; Quick Reference Guide	Email; training workshops; project newsletter	Fortnightly; weekly before go-live
Data Analytics Team	High	Requires data quality updates, forecasting model validation results and reporting/dashboard requirements.	Data Quality Report; Model Validation Report	Technical meeting; shared repository; Teams	Weekly during data and model phases
Finance Department	Medium	Requires budget usage, cost variance, procurement spending and approval-related financial information.	Budget Status Report; Cost Variance Report	Email report; finance review meeting	Monthly
QA & Change Management Lead	High	Requires defect status, test readiness, UAT progress, user training progress and adoption risks.	Test Report; Defect Log; Change Readiness Report	Jira; Teams; QA meetings	Weekly; daily during SIT/UAT
Cloud Services Provider	Medium	Requires cloud configuration requirements, SLA expectations, support requests and incident escalation information.	Cloud Configuration Document; SLA Report; Support Ticket Log	Vendor meeting; ticketing system; email	Fortnightly; ad-hoc for incidents
ML Solution Vendor	High	Requires model acceptance criteria, data	ML Requirements Specification;	Vendor meeting; Jira; email	Weekly during ML development

		requirements, technical constraints, performance feedback and issue resolution information.	Model Test Report; Issue Log		
ERP Software Vendor	Medium	Requires API specifications, integration schedules, testing windows, defect reports and change notices.	API Integration Specification; Integration Test Report	Technical meeting; vendor portal; email	Fortnightly; weekly during integration testing
System Integration Consultants	High	Require architecture decisions, integration dependencies, defect status, deployment readiness and cutover information.	Integration Plan; Defect Log; Deployment Checklist	Workshops; Teams; Jira	Weekly
Regulatory / Compliance Representatives	Low to Medium	Require evidence that privacy, security and compliance obligations are being addressed.	Compliance Report; Security Assessment	Formal documentation ; email	At compliance milestones

Change Management Plan

Objectives

In this project, "change" refers to the organizational, operational, behavioural, and technological transition that comes from implementing the Smart Retail Inventory and Demand Prediction System. The project does not just introduce a new technical platform; it also changes how inventory decisions get made, how procurement staff interact with demand forecasts, how Store Managers monitor stock risks, and how that legacy ERP, POS, and WMS data is used in the daily operations. Because of this, change management is needed to make sure the organization can move from a reactive, rule-based way of managing inventory over to a predictive, data-driven kind of inventory operation.

The purpose with this Change Management Plan is to provide a structured approach for preparing the stakeholders, supporting adoption, reducing resistance, and sustaining the benefits that come from the new system. This aligns with the principle that successful IT change depends not only on delivering the technical solution, but also on making sure the users understand the new way of working, accept it, and consistently use it. For this project, effective change management is particularly important because the system introduces machine learning-based forecasting, automated replenishment recommendations, and a Human-in-the-Loop procurement console. These features could end up altering existing responsibilities and will require users to trust the algorithmic recommendations while still keeping appropriate human oversight in place. The plan therefore focuses on communication, training, readiness assessment, support channels, resistance management, and post-go-live adoption monitoring.

Scope

The scope of this Change Management Plan covers all of the change activities that are needed to support a successful adoption of the Smart Retail Inventory and Demand Prediction System across the target organization. It applies to the 45 Store Managers, the central Inventory and Procurement Team, the IT support staff, the Data Analytics Team, and any relevant managers who will end up using the new system, supporting it, or making decisions based on what it produces. It also covers some indirect stakeholder groups like Finance, external vendors, and the Project Sponsor, but only in those cases where their involvement is needed for decision-making, communication, or operational readiness activities.

The plan includes stakeholder readiness assessment, change communication, user training, support arrangements, adoption monitoring, and also resistance mitigation both before go-live and after go-live. It also includes the preparation of change enablement materials, and examples of those would be Quick Reference Guides, training modules, workshop materials, FAQs, and post-deployment support procedures. This is consistent with what the scope plan says, where a Change Management and User Training Asset Library is identified as a formal deliverable that is designed to support that target adoption rate of at least 90% among the operational staff. However, this plan does not replace the formal technical change control that happens for scope changes, cost changes, schedule changes, or architecture changes. Any technical changes that affect the approved project baseline will continue to follow the project governance process and the integrated change control procedure. To give an example, changes that exceed AUD \$20,000, changes that alter the project scope, or changes that shift the go-live date by more than two weeks would all require formal Sponsor approval through that Tier 3 governance process. Because of this, the plan focuses on the people-side of adoption and the operational transition, while the technical baseline changes remain governed by the Change Control Log, the Project Manager, the Technical Lead, and the Steering Committee.

Impact Assessment

The current state of IT in the organization is characterized by a fragmented, reactive, and rule-based way of managing inventory. The organization currently runs across 45 retail locations and depends on siloed ERP, POS, and WMS systems, and that setup limits real-time visibility and ends up contributing to stockouts, overstocking, and procurement decisions that are not very efficient. Inventory decisions are mostly based on historical reports, manual judgment, and static replenishment thresholds rather than any kind of predictive analytics.

The future state is going to be a cloud-native, AI-enabled inventory and demand prediction environment. The system will bring together POS, WMS, and ERP data through secure APIs, generate SKU-level demand forecasts using those LSTM and XGBoost models, support automated purchase order generation, and also provide a centralized operational dashboard that comes with predictive alerts and Human-in-the-Loop approval functions. This setup creates several major areas of impact for the organization. One thing is business processes will change from a manual and retrospective inventory review over to a proactive, forecast-driven kind of replenishment. Procurement staff will end up spending less time calculating routine order quantities and more time reviewing exceptions, validating forecasts, and managing supplier constraints.

Another thing is roles and responsibilities are going to change. Store Managers will be expected to monitor those dashboard alerts, validate store-level stock risks, and provide feedback on local demand conditions, while the Inventory and Procurement Team will become the primary users of that HITL console. The QA and Change Management Lead will also need to coordinate training and adoption activities, just like what was already defined in the governance plan. A third aspect is technology adoption will require users to understand AI-generated forecasts, dashboard indicators, system alerts, and approval workflows. The main challenge here is not simply about giving people technical access, but rather about building user confidence in those algorithmic recommendations.

Finally, organizational culture is going to shift toward a more data-driven way of making decisions. This could create some tension for staff members who are used to doing experience-based ordering, or for those who might fear that automation will reduce the amount of decision authority they currently have.

Training and Support Plan

Training will be role-based and delivered before go-live, and the purpose of this is to make sure all affected users can perform the tasks they are required to do in the new system. Store Managers will get practical training on dashboard navigation, inventory alerts, stockout and overstock risk interpretation, and also the escalation procedures they need to follow. The Inventory and Procurement Team will receive a deeper kind of training on forecast review, those HITL approval workflows, automated purchase order generation, exception handling, and override documentation. IT support staff will get training on access management, incident logging, troubleshooting, and integration monitoring as well.

Training will be delivered through half-day workshops, guided demonstrations, scenario-based exercises, Quick Reference Guides, and some online learning materials. This is consistent with what the project schedule says, where half-day training workshops and Quick Reference Guides are included as activities before go-live. During that post-deployment Hypercare period, support will be provided to resolve any production issues that come up, answer user questions, and reinforce the correct ways of using the system. Support channels will include a Service Desk ticketing process, a dedicated Microsoft Teams support channel, FAQs, knowledge base articles, and then escalation up to the Technical Lead or the QA and Change Management Lead for any issues that are more complex.

Resistance Management Strategy

The main sources of resistance are expected to come from the Store Managers and also the procurement staff. Store Managers might worry that the system will end up disrupting their existing store-level routines or reduce the flexibility they have when making local decisions. Procurement staff could be concerned that automated replenishment will replace their manual judgment or reduce how important their role feels to them. Some users may also distrust those AI-generated forecasts, especially in cases where the recommendations conflict with their own experience or when demand patterns get affected by local events that the model might not see.

To mitigate these risks, the project will use early engagement, transparent communication, and practical involvement during UAT. Users will be shown that the system actually supports decision-making rather than fully replacing human judgment. That Human-in-the-Loop console will be emphasized as a control mechanism, one that lets authorized staff review the recommendations, adjust them when needed, and approve them before anything gets executed. Change champions will be selected from the Inventory and Procurement Team and from the Store Manager group, and their job will be to provide peer support and collect feedback from their colleagues. Resistance will also be monitored through training attendance, UAT feedback, support tickets, adoption dashboards, and post-go-live surveys. Where resistance is linked to a misunderstanding of how the system works, additional training will be provided; where it is linked to workflow concerns instead, the issue will be escalated through that change and governance process we have set up.

Monitoring and Review

Adoption and change success will be monitored using both quantitative KPIs and qualitative ones as well. The primary adoption KPI is that target system usage rate, which needs to be at least 90% across all intended users within 60 days after launch, and this aligns with the project's business value and adoption objectives that were set earlier. Some additional KPIs include the training completion rate, the user satisfaction score, the number of support tickets that remain unresolved, the average incident response time, the number of manual override actions being taken, the dashboard login frequency, and also the percentage of procurement decisions that get processed through the new system.

Operational KPIs will also be reviewed from time to time, and the purpose of that is to determine whether the change is actually delivering the business value we expected. These operational KPIs include forecast accuracy, how much stockout reduction we achieve, how much excess inventory holding cost gets reduced, API reliability, and system uptime. Review activities will happen on a weekly basis during that post-deployment Hypercare period, and then on a monthly basis during the first three months after deployment. The Project Manager, the QA and Change Management Lead, the IT Department, and the Inventory and Procurement Team will all review the adoption data, look at support trends, and examine user feedback. Any lessons learned from this process will be documented and then used to update the training materials, update the support procedures, and also inform future change initiatives down the line.

Project Monitoring and Control Plan

Introduction

Project monitoring and control is critical for that Smart Retail Inventory and Demand Prediction System because the project involves quite a few interdependent workstreams, and those include cloud infrastructure, historical data migration, AI/ML forecasting model development, API integration with those ERP, POS, and WMS systems, testing activities, user training, and also go-live support. In IT project management, monitoring and controlling is what makes sure the project stays aligned with its approved scope, its schedule, its cost targets, its quality standards, and the expectations of stakeholders throughout the whole execution phase. This is particularly important for this project because any delivery delays or uncontrolled changes could end up affecting the planned go-live date, the integration readiness, user adoption, and also the business benefits we are after, like improved forecast accuracy and reduced stockouts.

This plan puts in place a structured approach for tracking project performance, identifying any variances against those approved baselines, escalating issues when they come up, and then implementing either corrective actions or preventive actions as needed. It also makes sure that any proposed change to scope, time, cost, quality, or resources gets assessed through that formal change control process, rather than being introduced in some informal way during the development work.

Objective

The main objective with this Project Monitoring and Control Plan is to provide a consistent kind of mechanism for measuring project performance and making sure the project stays on track against those approved baselines we have set up. Specifically, the plan aims to monitor progress against the Work Breakdown Structure, the project schedule, the approved budget, the quality targets we defined, the risk register, the issue log, and also the stakeholder engagement requirements that were documented earlier. This helps with early identification of any deviations before they end up turning into major delivery problems down the line.

The plan also aims to improve decision-making by making sure project performance data gets collected, analyzed, and reported on a regular basis to the Project Manager, the project team, and the Steering Committee. Since this project has some defined success criteria in place, and those include things like $\geq 85\%$ SKU-level forecasting accuracy, zero critical data loss, $\geq 99.5\%$ uptime, and $\geq 90\%$ user adoption, the monitoring activities we do must connect the technical progress with those measurable business outcomes we are trying to achieve. Finally, the plan makes sure that any corrective actions, preventive actions, and formal change requests all get handled in an integrated way across scope management, schedule management, cost management, quality management, risk management, and stakeholder management as well.

Scope

The monitoring and control scope covers all of the major dimensions of project performance, and there is particular attention paid to those triple constraints of scope, time, and cost. Scope monitoring will make sure all of the approved deliverables get developed according to that approved scope baseline, and those deliverables include the AI demand forecasting module, the cloud data ingestion pipeline, the API middleware, the operational dashboard, the automated replenishment engine, the testing documentation,

the UAT sign-off, and the training materials. Any additional feature request that comes up, any change to the acceptance criteria, or any modification to system functionality will be assessed through that formal change control process so we can prevent scope creep from happening.

Schedule monitoring will track progress against the approved milestone plan, and that includes the cloud and data infrastructure establishment, the AI model and API middleware completion, the SIT sign-off, the UAT sign-off, the go-live date, the staff training activities, and the project closure work. Cost monitoring will compare the actual expenditure against that AUD 500,000 approved budget, and the budget includes allocated funding for system design, AI/ML model development, integration work, cloud infrastructure, testing, project management, training, change management, and also the contingency reserve we set aside. Quality monitoring will focus on test outcomes, defect trends, forecast accuracy, API latency, data integrity, uptime readiness, and UAT acceptance. The plan also monitors risks, issues, resource workloads, stakeholder communication, vendor performance, and user adoption readiness.

Monitoring Methods and Tools

The project will use a combination of schedule monitoring tools, cost monitoring tools, scope monitoring tools, quality monitoring tools, risk monitoring tools, and communication monitoring tools to provide a complete view of how the project is performing. For schedule monitoring, a Gantt chart will be used to compare the planned activities against what has actually been completed. This is appropriate because the schedule management lecture identifies Gantt charts as a key tool for visualizing project timelines, tracking progress against that baseline schedule, and identifying any overlaps, leads, and lags that might be there. The Project Manager will update the schedule on a weekly basis, and the update will be based on completed tasks, delayed tasks, milestone readiness, and also the dependencies that exist between workstreams like data engineering, ML model development, API integration, SIT, UAT, and training.

For cost monitoring, the project will apply a lightweight version of Earned Value Management or EVM. Those Planned Value (PV), Earned Value (EV), and Actual Cost (AC) metrics will be used to calculate Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), and Schedule Performance Index (SPI). These indicators allow the project team to see whether the project is under budget or over budget, and also whether the work that has been completed is consistent with what the planned schedule said should be done by now. Because this is a hybrid IT project with some iterative AI/ML development work in it, EVM will be combined with sprint-level progress reporting, burn-up and burn-down tracking, and deliverable-based acceptance.

For scope monitoring, the Requirements Traceability Matrix (RTM), the WBS, the issue log, and the change control log will all be used to make sure every approved requirement gets mapped to a deliverable, gets mapped to a test case, and gets mapped to an acceptance result. This helps prevent scope creep and makes sure any changes to the requirements get formally reviewed before they are accepted. For quality monitoring, the QA and Change Management Lead will track defect density, those Severity 1 and Severity 2 defects, the SIT pass rate, UAT completion, API latency, data migration accuracy, forecast model performance, and also the training completion rate. Risk and issue monitoring will be conducted through those weekly risk register reviews, through issue log updates, and through escalation reports. Communication monitoring will be supported by weekly team stand-ups, fortnightly project status reports, Jira and Confluence dashboards, and monthly Steering Committee reviews.

Control Mechanisms

Control mechanisms will be applied whenever those monitoring activities end up identifying a significant kind of deviation from the approved project baseline. The Project Manager will first figure out whether

the deviation relates to scope, schedule, cost, quality, resources, risk, stakeholder engagement, or procurement. Minor deviations that can be resolved within the approved project tolerances will be managed directly by whatever workstream owner is responsible for that area, and then they will be reported in the weekly status update. More significant deviations, especially those that affect approved milestones, budget limits, critical deliverables, security compliance, UAT sign-off, or go-live readiness, will be escalated up to the Steering Committee for their attention.

The control process will follow a structured sequence from start to finish: first you identify the variance, then you analyze the root cause behind it, then you assess what kind of impact it has, then you define the corrective actions or preventive actions that need to be taken, then you assign an owner to those actions, then you agree on a due date for completion, then you implement the response, and finally you verify whether the issue has actually been resolved.

Corrective Actions

Corrective actions will be used whenever the actual project performance has already deviated from that approved baseline, and the project needs to be brought back onto the right track. To give an example, if that API integration work falls behind schedule, the Project Manager might reassign some developers to it, prioritize those critical integration endpoints, or approve a limited amount of schedule compression by means of fast-tracking. If cloud infrastructure spending ends up exceeding what was planned in the budget, the Technical Lead will review the instance usage, remove any resources that are not being used, and optimize that GPU consumption or storage consumption. If SIT ends up identifying some Severity 1 or Severity 2 defects, the QA Lead will create a defect recovery plan, assign fix owners to each defect, and require retesting before UAT is allowed to begin. Any corrective actions that get taken must be recorded in that issue log and then reviewed repeatedly until they are finally closed out.

Preventive Actions

Preventive actions will be used to reduce the probability of any future deviations before those deviations actually happen. For this project, preventive actions include things like weekly schedule variance reviews, early updates to the risk register, peer reviews of technical designs, regular data quality checks, cloud cost alerts, test readiness reviews, and stakeholder feedback sessions that happen before UAT begins. Because AI/ML work can be uncertain due to data quality issues, model tuning challenges, and forecasting accuracy risks, the project team will use early model validation checkpoints rather than just waiting until that final testing phase to confirm how well the model performs. Preventive actions will also include maintaining that contingency reserve we set aside, monitoring resource workloads, confirming vendor availability ahead of time, and making sure the training materials are prepared before go-live so we can reduce those adoption risks as much as possible.

Change Control Integration

Project monitoring and control will be directly integrated with that Change Management Plan we set up earlier. Any variance that cannot be corrected within the approved baseline will trigger a formal change request that needs to go through the proper process. This includes changes to project scope, delivery schedule, budget allocation, acceptance criteria, technical architecture, data integration requirements, security controls, training approach, or the go-live timing. The change request has to document the reason for wanting the change, the deliverables that would be affected, the impact on scope, schedule, cost, quality, risks, and stakeholders, and also the recommended approval pathway that should be followed.

Minor operational adjustments may be approved by the Project Manager if they do not end up affecting those baselines we have in place. However, changes that have strategic implications, financial implications, contractual implications, or implications for the go-live date must be escalated up to the Steering Committee or the Project Sponsor for their review and approval. Once a change gets approved, it will be recorded in that Change Control Log, and then the relevant project documents will be updated, and those documents include the schedule, the budget, the risk register, the issue log, the RTM, and the communication plan. This makes sure the monitoring results are not treated as isolated observations that stand alone, but are instead converted into controlled decisions that help preserve project governance and delivery accountability.

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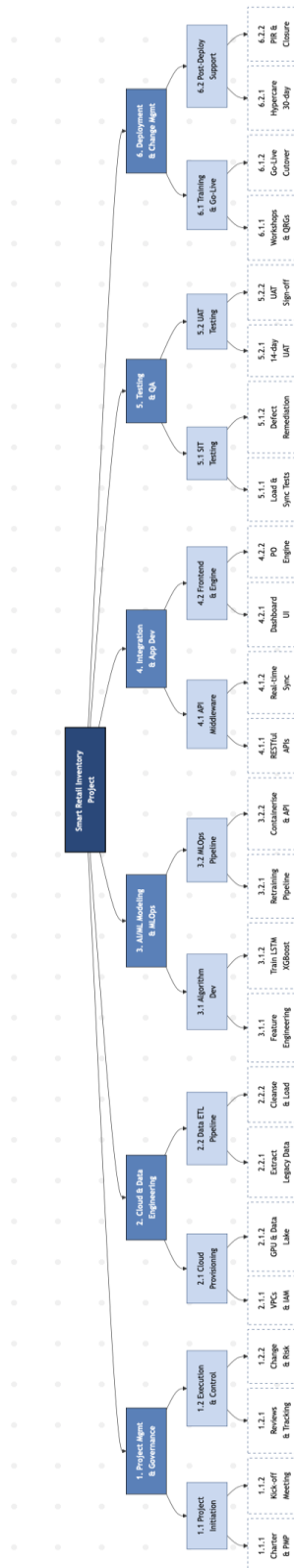
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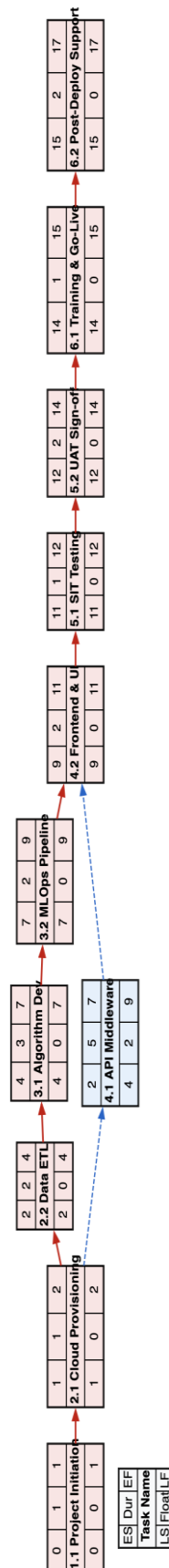
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Appendix

The Work Breakdown Structure



Network Diagram



Gantt Chart

